Grey Box: A Closed Model's Answers Reveal a Latent Geometry It Shares with Open Models, and Open Models Can Monitor It

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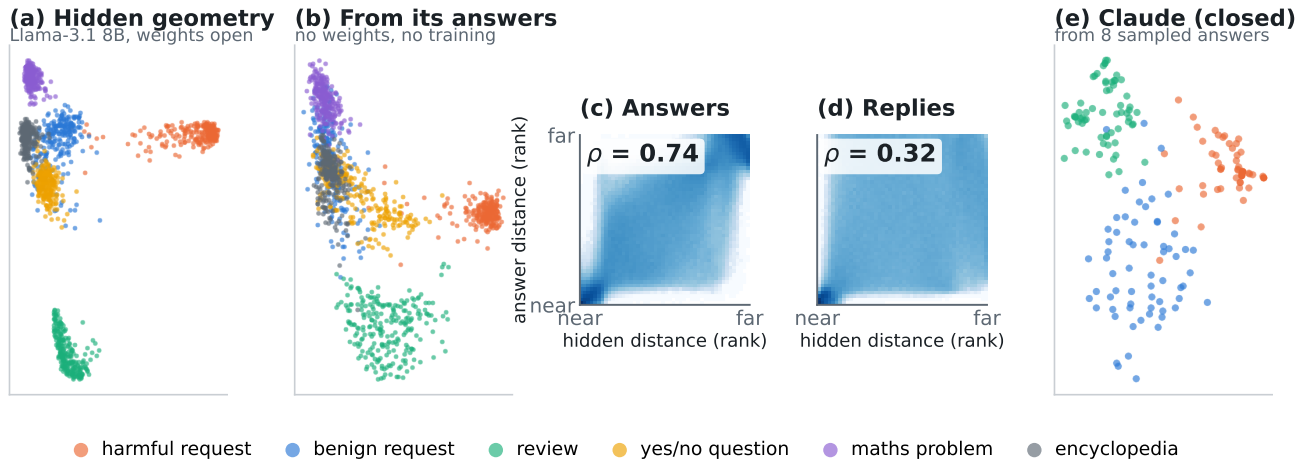


Figure 1. **A model's latent geometry, read from its answers.** (a) The hidden geometry of an open model (classical scaling of its hidden states on 1,440 held-out contexts). (b) The same contexts placed only by the model's yes/no answers, in the exact Fisher–Rao coordinates of Theorem 1; no weights are used and nothing is trained (rotated onto (a) for display). (c, d) Every pair of contexts: rank of the hidden distance against the rank of the distance from the answers, or from the model's own sampled replies; Spearman ρ is the correlation of each picture (over eight models: answers 0.41–0.74, replies 0.22–0.44, input text 0.22–0.30). (e) The same map for a closed model, Claude Haiku 5.5, from eight sampled answers per question.

Abstract

The hidden coordinates of a closed model cannot be read from its outputs: a gauge symmetry of the network leaves every output unchanged. What can be read is a geometry. Asking yes/no questions in side branches of the conversation gives answer laws whose Fisher–Rao distances are exactly Euclidean in arcsine coordinates, and a cross-fit estimator is unbiased from sampled text alone. We measure this observable geometry on eight open models and test it against their hidden states, with every test preregistered and its lock published before the data. It tracks the hidden geometry (Spearman 0.41–0.74) far better than the input text (0.22–0.30) or the model's own replies (0.22–0.44). But it is not the model's own: on 1,000 fresh contexts, a model's answers match other models' hidden geometries as well as its own (own-model excess +0.002), on new kinds of context too, and another open model's hidden state is the better stand-in. What a closed model's answers reveal is a latent geometry it shares with open models; Claude Haiku 5.5 reveals the same. This is a measured reason why an open model can monitor a closed agent: reading transcripts through an open model, a monitor trained only on open agents catches 94% and 98% of two closed agents' violations at a 5% label-free review budget, and cuts the episodes with an executed violation from about 7% to 0.2% live. The model-specific final geometry is recoverable exactly only with full log-probabilities. We release the tool.

1. Introduction

Interpretability reads what a model knows and what it will do from its hidden state (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025), and for agents a probe can stop an unsafe action before it runs (Fayyazi et al., 2026).

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These tools need the weights. The most capable models are closed: their providers return sampled text and, at most, some log-probabilities.

What can an outside observer know about a closed model's state? Not its coordinates: any reparametrisation of the hidden space that the next layers undo leaves every output unchanged, a gauge symmetry (Proposition 1). Distances between states survive this symmetry, so we ask which geometry the outputs reveal. Ask the model yes/no questions about a context, each in its own side branch. In the angles $\vartheta = 2\arcsin\sqrt{P(\text{YES})}$, the Fisher–Rao distance between two contexts' answer laws is plain Euclidean distance (Theorem 1): a change of variables, not a learned embedding, and estimable without bias from sampled text on any chat API.

What we find. This observable geometry tracks a model's hidden geometry far better than the input text or the model's own replies. But a control that an independent review asked for shows that it is not the model's own: a model's answers match *other* models' hidden geometries exactly as well as its own, on fresh contexts and new kinds of context, and an open model's hidden state is a better stand-in than the model's own answers. What outputs reveal is the latent geometry that models share (Picozzi, 2026; Huh et al., 2024). A closed model, Claude, reveals the same. This turns a common assumption into a measurement: an open model that reads the same tokens is a principled stand-in for a closed one. We use it to monitor closed agents, and we show that the model-specific part of the geometry needs full log-probabilities (Theorem 2).

Contributions.

1. *Theory* (Section 2): the observable geometry of a model's state in exact flat coordinates, its invariance, its link to the hidden state, and an exactly unbiased estimator from sampled answers; the exact final geometry from full log-probabilities, after Carlini et al. (2024).
2. *Measurement* (Section 3): on eight open models, the answer geometry tracks hidden geometry better than the input text and the model's replies; it is shared across models, with no own-model excess, on fresh and new kinds of context; Claude reveals the same geometry. We report where the answers lose.
3. *Application to closed agents* (Section 4): with the transcript and the action sufficient for policy checks (Corollary 1), a monitor that reads them through an open model predicts and prevents two closed agents' violations live, also when the agent knows it is monitored.
4. *A released tool* (Section 5): one command measures a model's observable geometry; an SDK, a server and a browser trainer monitor agents.

Every confirmatory test was preregistered: its document and analysis code were hash-locked (SHA-256) before the data existed, and for the later tests, all geometry tests included, the lock was published first. Tests that failed, a dropped design, a correction of our own theorem, and the limits of each claim are in Sections A and J.

2. Theory

2.1. Setting

A language model reads a context x (tokens) and reaches an internal state $s(x)$, a deterministic function of the tokens. An outside observer sees only outputs: sampled text and, on some APIs, the log-probabilities of the next token. A *battery* $Q = (q_1, \dots, q_m)$ of yes/no questions is asked in side branches: each question is appended to x in its own call, so the model's real conversation never sees it, and the answers are independent given x. The *interrogation state* is $\Psi_Q(x) = \big(P(\text{YES} \mid x \oplus q_j)\big)_j$, and the answers follow the product law $P_x = \bigotimes_j \text{Bernoulli}(\Psi_{q_j}(x))$. Where a model is open, its hidden state $h(x) \in \mathbb{R}^d$ at a layer is the ground truth that we test against.

2.2. What cannot be read: a gauge symmetry

Proposition 1 (gauge)**.** *Take a pre-norm residual network with RMS normalisation, with the normalisation gains folded into the adjacent weights. For every orthogonal G on the residual stream, the network whose embedding, write matrices and write biases are multiplied by G on the left, and whose read matrices and unembedding are multiplied by $G^\top$ on the right, has hidden states Gh and the same output distribution on every input. At a layer that is read only through linear maps, the same holds for every invertible G, with G^{-1} in place of $G^\top$.*

No estimator that uses only outputs can recover hidden *coordinates* beyond this group (computational invariance; Ashkboos et al., 2024). Distances and angles between states are invariant under the orthogonal group, so the *geometry* of the states is the natural target. Post-block normalisation (Gemma-3, OLMo-3) restricts G to the maps that commute with its gains (Section B).

2.3. What can be read: the observable geometry

Let $d_Q(x, x')$ be the Fisher–Rao geodesic distance between the answer laws P_x and $P_{x'}$.

Theorem 1 (observable geometry)**.** *(a)* Exact flat coordinates. *With* $\vartheta_j = 2\arcsin\sqrt{\Psi_{q_j}}$, $d_Q(x,x')^2 = \sum_j \big(\vartheta_j(x) - \vartheta_j(x')\big)^2$. *(b)* Invariance and uniqueness. *d_Q depends on the outputs only, so it is invariant under the gauge group. Up to a constant factor, the Fisher metric is the only Riemannian metric on the probability simplex over $\{0,1\}^m$ that is invariant under Markov morphisms*

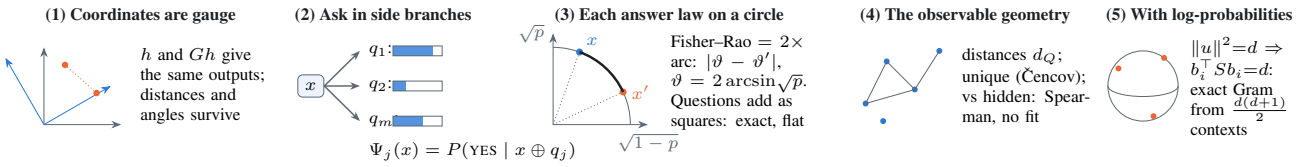


Figure 2. **The method.** (1) Hidden coordinates are not observable (Proposition 1). (2) Yes/no questions are asked in side branches; the real conversation never sees them. (3) A Bernoulli law p is the point $(\sqrt{1-p}, \sqrt{p})$ on a quarter circle, and its Fisher–Rao distance to another is twice the arc between them; independent questions add as squares, so the angles are exact flat coordinates (Theorem 1). (4) The distances between contexts are the observable geometry; we compare it with hidden geometry without fitting anything. (5) With full log-probabilities, the normalisation sphere gives a linear system for the exact Gram matrix of the final states (Theorem 2).

(Čencov); d_Q is the geodesic distance it induces on the submanifold of independent answer laws (in the whole simplex the distance is $2\arccos\prod_j \cos\frac{\vartheta_j-\vartheta'_j}{2}$). (c) Relation to the hidden state. *If $\Psi_j = \sigma(a_j(s))$ with smooth a_j, then for nearby states $d_Q^2 = \Delta^\top G_Q(s)\,\Delta + O(|\Delta|^3)$, with the pulled-back Fisher metric $G_Q = \sum_j \Psi_j(1-\Psi_j)\nabla a_j \nabla a_j^\top$. (d)* Estimation from sampled answers. *Let $\mu_n(p) = \mathbb{E}\,\hat\vartheta$ for the Anscombe angle $\hat\vartheta = 2\arcsin\sqrt{(k+3/8)/(n+3/4)}$, $k \sim \mathrm{Bin}(n,p)$; μ_n is known, strictly increasing, and $\mu_n \to \vartheta$. For two independent halves A, B of n samples each, $\hat D^2(x,x') = \sum_j \big(\hat\vartheta_j^A(x) - \hat\vartheta_j^A(x')\big)\big(\hat\vartheta_j^B(x) - \hat\vartheta_j^B(x')\big)$ satisfies $\mathbb{E}\hat D^2 = \sum_j \big(\mu_n(\Psi_j(x)) - \mu_n(\Psi_j(x'))\big)^2$ exactly, for every Ψ, when every cell has the same n.*

Part (a) is a change of variables: no model is trained and nothing is fit. Classical scaling of d_Q gives coordinates that are exact up to a rigid motion. Part (b) makes d_Q the canonical output geometry of the answers; Picozzi (2026) studies the same object for next-token distributions. Part (c) links it to the hidden state; since s is the whole cache that the questions read, its predictions for one layer's vector are heuristic: agreement should rise with the number of questions and saturate as G_Q gains rank. Part (d) makes the geometry measurable on any chat API, at $m \cdot n$ calls of one output token per context; the familiar rule of subtracting the large-n variance $1/(n+\frac{1}{2})$ is not exact, because near $\Psi \in \{0,1\}$ the variance is smaller and the bias is $\cot\vartheta/(4n)$, and an answer that is neither yes nor no, or clipping negative estimates at 0, breaks the exactness of (d) (Section B). The answer law is conditional on a yes or no answer.

Where an API returns full log-probabilities, the final state is not estimated but recovered, by the construction of Carlini et al. (2024, App. H).

Theorem 2 (exact recovery; after Carlini et al.)**.** *Let the logits be Wu, with $u = x/\mathrm{rms}(x)$ the normalised final state, the normalisation gain folded into W, and no output bias. The centred log-probabilities $c(x)$ lie in a d-dimensional subspace; write $b = Q_d^\top c$ for an orthonormal basis Q_d of it. Then $u = Kb$ for an unknown invertible K, and $\|u\|^2 = d$ gives $b_i^\top S b_i = d$ with $S = K^\top K$: a linear system in S. If the $b_i b_i^\top$ span the symmetric matrices (generically, $N \geq d(d+1)/2$ contexts), S is unique, and the Gram matrix of the final states, $b_i^\top S b_j = u_i^\top u_j$, is exact for every new context. The states follow up to an orthogonal map: the gauge group.*

The subspace (Finlayson et al., 2024) and the ellipsoid fit that removes all but an orthogonal ambiguity (Carlini et al., 2024, App. H) are known; we state the consequence for the gauge-invariant geometry and test its sharp transition at $d(d+1)/2$ contexts. Recovery holds up to the normalisation's ϵ, and near the threshold the system is badly conditioned.

2.4. From the geometry to monitoring agents

An agent is a model that acts through tools. At step t the operator sees the visible tokens V_t (instructions, task, earlier calls and their results) and the proposed action A_t; $Y \in \{0,1\}$ marks a policy violation, decided by A_t, V_t and facts W of the world. A *monitor* scores (V_t, A_t) and escalates the action when the score is at or above a cut c; its *budget* β is the share of actions that it escalates.

Corollary 1 (sufficiency of the proposed action)**.** *Let $Y = g(A_t, V_t, W)$ with $W \perp\!\!\!\perp Z_t^- \mid V_t$, where Z_t^- are all tokens before the action, hidden ones included. Then $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$, and every monitor that reads the agent's internals at or before the action has a Bayes risk at least that of the best monitor that reads (V_t, A_t).*

For policy labels, a monitor therefore needs the transcript and the action, not the weights. Our monitor is a *reader*: an open model reads the same visible tokens and the action, and a linear probe on its hidden state, trained on labelled actions of open agents, scores the action. With reasoning off, the agent's state is a function of tokens that the reader also reads, so the reader's state can also replicate the agent's state (Propositions 5 and 6, Section C).

Proposition 2 (a frozen cut)**.** *Let the scores of n unlabelled warm-up actions and of a new action be exchangeable with no ties, and let $\hat c$ be the k-th smallest warm-up score. Then $P(S_{n+1} > \hat c) = (n+1-k)/(n+1)$, which for $k = \lceil (1-\beta)(n+1) \rceil$ lies in $\big(\beta - \frac{1}{n+1}, \beta\big]$.*

Proposition 3 (shift and bursts)**.** *If deployment scores follow Q and warm-up scores follow P, a frozen cut escalates within $K(P,Q) + \frac{2}{n+1}$ of β, with K the Kolmogorov distance. A cut set at the top β of a window in which a share $\rho > \beta$ of the actions are violations catches at most β/ρ of them; a frozen cut catches each violation with probability $P(S \geq \hat{c} \mid Y = 1)$, whatever ρ is.*

A cut that keeps moving with recent traffic thus fails exactly when an attack arrives; a frozen cut keeps its rate, and after a shift of the safe traffic (a new prompt, a new agent) it must be set again on the new unlabelled traffic. The full statements, the monitor's algorithm and a ceiling on robustness to rewording (Proposition 7) are in Section C.

Metrics. For geometries: Spearman correlation between two distance matrices over all pairs of contexts (no fitted parameter). For monitors: the AUROC within strata (template, tool or suite), and the share of violations caught when exactly β of the actions are escalated. Intervals are 95% percentile intervals from a bootstrap over contexts, scenarios or episodes. Proofs are in Section B.

3. What a model's answers reveal: results

Data. 4,800 contexts of six kinds (harmful and benign requests, reviews, yes/no questions, maths problems, encyclopedia text), split into 3,360 development and 1,440 confirmation contexts, and 1,000 fresh contexts for the last test, 400 of them of four kinds never used before (news, programming tasks, jailbreak behaviours, prompt injections). A battery of 88 questions about the message (topic, intent, tone, demands, harm, and the model's own stance). Eight open chat models from six families. Ground truth: each model's residual stream at the last prompt token, at the layer chosen on development contexts only. All results are on held-out contexts, as preregistered.

The answers track the hidden geometry; the input text and the replies do not. On every model the answer geometry matches the hidden geometry better than the geometry of the input text (Table 1; gains +0.16 to +0.49, every lower bound at least 0.141; preregistered, 8 of 8). The model's own replies, the other output that an API gives, do little better than the text (0.22–0.44; preregistered: H1 passes, H2 passes, H3 passes): one reply and the mean of eight differ by at most 0.031, while from 8 to 88 questions the answers' agreement rises by +0.02 to +0.25 (Figure 3b). Answers sampled from the model's own probabilities, 8 per question, keep 97–100% of the exact agreement. Two qualifications. Much of the agreement is the separation of kinds of context: within one kind it is weaker (median 0.29 for answers, 0.17 for text; answers ahead in 33 of 48 model–kind cells; exploratory). And on the fresh contexts of the last test agreement is lower (answers 0.22–0.64, text 0.15–0.23, replies 0.16–0.43); there Gemma's replies beat its answers.

Table 1. **Which observable carries the state?** Spearman correlation of each geometry with the model's own hidden geometry, over all pairs of 1,440 held-out contexts; no parameter is fit. Answers: Fisher–Rao distances over 88 questions, exact (log-probabilities) or from 8 answers sampled from the model's own probabilities. Replies: mean embedding of 8 sampled replies. Text: embedding of the input. Last column: answers − text, 95% interval.

	Answers				
Model	exact	8 samples	**Replies**	**Text**	**answers − text**
Qwen2.5 7B	**0.74**	0.74	0.30	0.28	+0.46 [0.44, 0.48]
Llama-3.1 8B	**0.74**	0.73	0.32	0.26	+0.48 [0.46, 0.50]
Mistral 7B	**0.70**	0.70	0.22	0.22	+0.49 [0.47, 0.51]
Qwen3 8B	**0.66**	0.65	0.33	0.30	+0.36 [0.34, 0.39]
Phi-4 mini	**0.58**	0.57	0.26	0.28	+0.30 [0.28, 0.33]
OLMo-3 7B	**0.57**	0.55	0.34	0.29	+0.28 [0.26, 0.30]
Gemma-3 4B	**0.56**	0.56	0.44	0.28	+0.28 [0.26, 0.31]
gpt-oss 20B	**0.41**	0.40	0.25	0.25	+0.16 [0.14, 0.19]

The geometry is shared, not the model's own. An independent review asked whether a model's answers match *its own* hidden geometry better than another model's. They do not. On 1,000 fresh contexts, the agreement of model A's answers with model B's hidden geometry has no own-model excess beyond row and column effects: +0.002 [-0.001, +0.005], inside the preregistered equivalence band of ± 0.02 (Figure 3d); the same held on the earlier contexts and in every category of question, including the questions about the model's own stance (exploratory). The hidden geometries of different models agree with each other (0.18–0.80), and an open model's hidden state (Qwen3-8B) matches another model's hidden geometry better than that model's own answers do, for 6 of 7 models (-0.05 to +0.36; preregistered). So what a model's outputs reveal of its state is the geometry that models share, as Picozzi (2026) found for next-token distributions and in line with converging representations (Huh et al., 2024); the model-specific remainder of the hidden geometry is invisible in its answers. Schiekiera et al. (2026) report behavioural signal beyond cross-model consensus for word-similarity judgements; with our battery we find none.

It holds on new kinds of context. On the 400 contexts of the four new kinds, the answers beat the text for 6 of 8 models (0.28–0.60 vs 0.21–0.33; preregistered; for Gemma and gpt-oss the interval includes 0).

A closed model. Claude Haiku 5.5 answered 24 questions, 8 times each, on 192 held-out contexts. Two independent halves of the samples give the same geometry: Spearman 0.95 [0.94, 0.96], above the bound of 0.7 fixed in advance from open models, and +0.44 [0.39, 0.49] beyond the text geometry (preregistered). With the cross-fit estimator the geometry is close to Euclidean (negative eigenvalue mass 2%, against

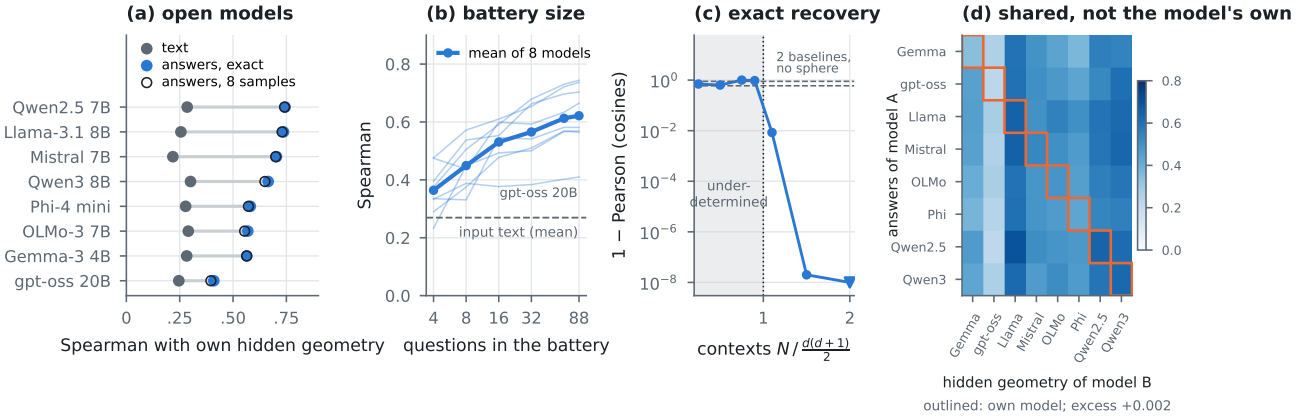


Figure 3. **The observable geometry, preregistered results.** (a) Agreement of each open model's answer geometry with its own hidden geometry, exact and from 8 sampled answers, against the geometry of the input text (1,440 held-out contexts). (b) Agreement against the number of questions. (c) Exact recovery of the final hidden geometry from log-probabilities (Qwen3-0.6B, 2,000 held-out contexts) against the number of contexts; the threshold is $d(d+1)/2$. (d) Agreement of model A's answer geometry with model B's hidden geometry on 1,000 fresh contexts; the own-model diagonal (outlined) is not special.

15% with the variance-subtraction rule). Most cells are unanimous, so this stability partly reflects how deterministic Claude is. Its agreement with the open models' hidden geometries is 0.45–0.51. With the same 24 questions and 8 samples, open models' answers agree with other open models' hidden geometries at 0.41–0.76 (median 0.58; exploratory), so the closed model reveals the shared geometry, at the lower end of the open models' range.

The model-specific part needs log-probabilities. The final hidden geometry of a model is its own, and Theorem 2 recovers it exactly from full log-probabilities. On Qwen3-0.6B ($d = 1{,}024$), from 1.15 million self-generated contexts, the recovered cosines of 2,000 held-out contexts correlate with the true ones at 0.00–0.36 below $d(d+1)/2$ contexts, at 0.9915 at $1.1\times$, and at $2\times$ the median error is 7×10^{-6} (maximum 1.5×10^{-4}) (Figure 3c; preregistered). Closed APIs do not return full log-probabilities.

Where answers lose. For detecting *that* a model's state moved (a hidden system prompt), sampled replies had more power than sampled answers at equal calls in every model and condition on development data, so we did not run that test (Section A).

4. Application: monitoring closed agents

Because the observable geometry is shared, an open model that reads the same tokens holds a measured stand-in for a closed model's state (Section 3, S2). We use this to monitor closed agents: the monitor reads the transcript and the proposed action through an open model, which Corollary 1 says is enough for policy checks. **Agents and monitors.** CAPE-Bench (Fayyazi et al., 2026) gives matched tool-use tasks in which one fact decides whether an action is authorised; an environment checker labels each action. Open agents: Gemma-4 (26B), gpt-oss-20b, Qwen3.8-27B and Muse-Glimmer-30B. Closed agents, through their public APIs with reasoning off: Claude Haiku 5.5 and GPT-6-luna. The reader is Qwen3-8B (layer 32), with a logistic probe trained only on 17,623 Gemma-4 steps; baselines are the agent's own internals (where open), TF-IDF with the same labels, a zero-shot 70B judge and the action's tool alone.

The transcript is enough. Corollary 1 says that the agent's internals cannot beat a monitor of the transcript and the action. A probe on Gemma-4's own internals scores 0.841 at the start of the turn and 0.979 after its reasoning; the reader probe, trained on the same labels, scores 0.994 ($+0.015$ [0.007, 0.026] over the internals) and is non-inferior on unseen task families (Table 2). Given the action, adding the agent's hidden reasoning changes the AUROC by -0.0014 [-0.0046, $+0.0014$] (exploratory). The reader's state also replicates the agent's: on four architectures it reconstructs 46–66% of the variance of the main components of the decision state on unseen scenarios, more than text features, and a probe trained on true states scores at least as well on the replica (Sections C and H). What the reader adds over word counts is robustness to shift: TF-IDF with the same labels ties it on the training benchmark but is at chance on AgentDojo (0.503 vs 0.781).

Two closed agents. Nothing was fit on the closed agents. At a cut set at the top 5% of the agent's own unlabelled traffic, the monitor catches 94.1% of Claude Haiku 5.5's 68 violations (zero-shot judge: 0%; TF-IDF: 63.8%) and 98.5% of GPT-6-luna's (judge 37.9%; TF-IDF 49.0%). The scale of the scores does not transfer between agents (on Claude a cut at probability 0.5 catches 3%), but the ranking does, so the cut must come from the monitored agent's traffic.

Table 2. **Detecting violations before the action runs.** Within-template AUROC (within suite on AgentDojo); nothing is fit on a test agent. The internals need the agent's weights; the other monitors read the transcript and the action. Bold: best; underline: second. Last column: the preregistered contrast (I: reader − internals; NI: non-inferiority to internals; J: reader − judge; J_{tool}: within tool), with its 95% interval. †Exploratory, on held-out scenarios; internals: a probe trained on the agent's own states and labels.

Setting	Agent	Actions	Monitor				Preregistered contrast
			Agent's internals	70B judge (zero-shot)	TF-IDF (same labels)	**Grey-box reader**	
Open agents: the training agent, then three other architectures							
Same scenarios	Gemma-4	12,128	0.979	0.910	<u>0.992</u>	**0.994**	I +0.015 [0.007, 0.026]
Unseen task families	Gemma-4	12,128	**0.976**	0.910	0.950	<u>0.969</u>	NI −0.007 [−0.014, 0.000]
Mixture of experts	gpt-oss-20b	5,229	–	0.833	**0.943**	<u>0.915</u>	J +0.082 [0.043, 0.129]
Hybrid attention†	Qwen3.8-27B	2,378	0.955	0.835	<u>0.966</u>	**0.972**	J +0.135 [0.035, 0.258]
Dense†	Muse-Glimmer	2,657	0.929	0.911	<u>0.960</u>	**0.968**	J +0.059 [0.017, 0.131]
Closed agents, through their APIs (reasoning off)							
CAPE-Bench	Claude Haiku 5.5	3,128	–	0.633	**0.997**	<u>0.991</u>	J +0.360 [0.302, 0.413]
CAPE-Bench	GPT-6-luna	3,627	–	0.942	<u>0.968</u>	**0.995**	J_{tool} +0.039 [0.015, 0.066]
A new benchmark: AgentDojo prompt injection, a policy absent from training							
No AgentDojo labels	two open	3,455	–	<u>0.537</u>	0.503	**0.781**	J +0.249 [0.136, 0.369]
Labels of open agents only	GPT-6-luna	3,925	–	<u>0.787</u>	–	**0.922**	J +0.136 [0.078, 0.191]

Table 3. **Live prevention** (preregistered; 500, 450, 450 and 400 scenarios, each without and with the gate). Episodes with an executed violation and episodes that reached the goal, in %; Δ: gated − ungated violation rate, with its 95% interval.

Agent, gate	Violations		Δ [95% CI]	Goals	
	none	gate		none	gate
GPT-6-luna, block	6.6	**0.2**	−.064 [−.088, −.044]	47.4	45.6
Claude, block	6.7	**0.2**	−.064 [−.087, −.044]	40.7	34.9
hold for review[a]	6.7	**0.0**	−.067 [−.091, −.044]	40.7	38.7
told, cut re-set[b]	6.5	**0.0**	−.065 [−.090, −.043]	38.5	35.5

[a]An oracle reviewer; same scenarios as the row above, run separately. [b]Block, with the cut set again on the told agent's unlabelled traffic: it escalated 4.7% [3.5, 6.1] of actions (budget 5%; the old cut 16.6%).

Frozen after 500 actions, it keeps catching 94% under bursts in which half the actions are violations, while a cut that moves over a window of 5,000 actions falls to 36%, as Proposition 3 predicts (Figure 6a).

Live prevention. We ran the monitor as a server in each closed agent's tool loop, on the same scenarios with and without the gate. In all four preregistered live tests the gate removed nearly all executed violations (Table 3 and Figure 4), and no agent executed a violation after a block. Holding an escalated action for a reviewer kept the safety gain; its preregistered bound on the goal cost was not shown (−0.020 [−0.051, +0.007]; Section A). The round trip was 89 ms on GPT-6-luna and 80 ms on Claude (median; 95th percentiles 215 and 139 ms).

An agent that knows it is monitored, and an attacker who queries the score. Told in its system prompt that a safety monitor reviews every tool call, Claude violated as often as before, and the monitor caught 97.1% of its violations at the frozen cut (83.6% when not told; +0.136 [0.050, 0.225]; preregistered). The notice moved all scores up, so the frozen cut escalated 15.2% of actions instead of 5%, within the Kolmogorov bound of Proposition 3 ($K = 0.24$); a cut set again on the told agent's unlabelled traffic escalated 4.7% live and still removed every executed violation (Table 3). An attacker who can query the score and rewrites only the free text of a violating call lowered the share caught from 96% to 29%; two preregistered defenses failed (51% and 60%), close to the ceiling of Proposition 7 (Section A). The score must stay hidden from the agent.

Labels of a new policy transfer from open agents. Prompt injection on AgentDojo is a policy absent from our training data. Labels of it collected on two *open* agents transfer to the *closed* GPT-6-luna: AUROC 0.922 against the judge's 0.787 (+0.136 [0.078, 0.191]; Table 2). The limit of a monitor is the coverage of its policies, not the agent.

5. The tool

We release a Python package and a website (links in the camera-ready version). One command, `graybox-geometry`, measures a chat model's observable geometry over the user's contexts: it reads $P(\text{YES})$ exactly where the API returns log-probabilities, and otherwise samples two independent halves and uses the cross-fit of Theorem 1(d); it reports the number of calls and a cost estimate before it runs, and its tests check each theorem on cases with a known answer. For agents, a user imports logs (OpenAI or Anthropic format), labels actions and trains a monitor in the browser, where the logs never

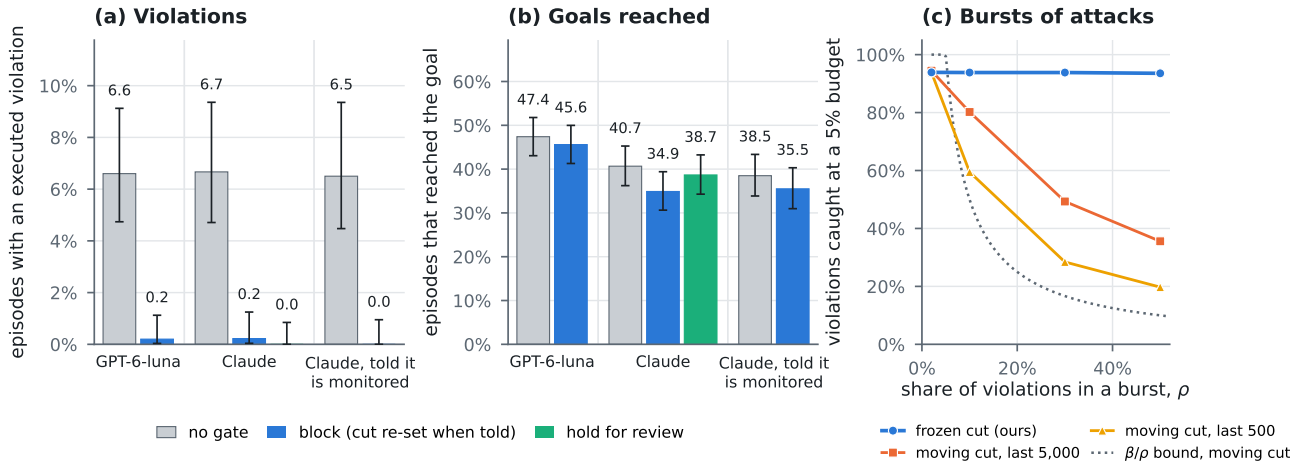


Figure 4. **Live prevention and bursts.** (a) Episodes with an executed violation and (b) episodes that reached the goal, in the four preregistered live tests: GPT-6-luna and Claude Haiku 5.5 without a gate and with blocking; Claude with hold for review (an oracle reviewer, run separately); and Claude told that it is monitored, with the cut set again on its unlabelled traffic. Bars: rates; whiskers: 95% Wilson intervals. (c) Violations caught at a 5% budget as the share ρ of violations in a burst grows, for the frozen cut and two cuts that move with recent traffic (Claude's scores, replayed); dotted: the β/ρ bound of Proposition 3.

leave the machine; a guard places the monitor in a tool loop, an MCP proxy or one of three agent frameworks, and sets the cut again on new unlabelled traffic after a change of prompt (Section F).

6. Related work

Information geometry and shared representations. The Fisher metric and its uniqueness are classical (Čencov, 1982; Campbell, 1986; Amari, 2016), as are the arcsine transform (Anscombe, 1948) and classical scaling (Torgerson, 1952); representational similarity analysis compares distance matrices without a fit (Kriegeskorte et al., 2008). Representations of different models converge (Huh et al., 2024). Picozzi (2026) show that the Fisher–Rao geometry of next-token distributions is set by behaviour up to symmetries and shared across architectures; Schiekiera et al. (2026) compare behavioural and hidden-state geometry in eight models. We measure the geometry through yes/no answers on sampled text, so it applies to closed models; we test it against hidden states with a cross-model control; and we use its sharing to justify monitoring closed agents through an open model. **Black-box access.** Logits span a d-dimensional image (Finlayson et al., 2024), and a production model's output projection is recovered up to an orthogonal map (Carlini et al., 2024); Theorem 2 uses the same structure for the gauge-invariant hidden geometry. Replies serve to test whether an API serves a given model (Gao et al., 2025); in our data they are the more sensitive output for a change and the less faithful one for the state. Answers to questions serve as interpretable embeddings (Benara et al., 2024), detect lies with follow-up questions (Pacchiardi et al., 2024; Cooney et al., 2026; Sam et al., 2025), and forecast behaviour (Levy et al., 2026); open readers probe closed transcripts (conti et al., 2026). **Probes, monitors and control.** Probes on internals read truth, refusal and deception (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025), and CAPE probes an agent before each action (Fayyazi et al., 2026). Chain-of-thought monitors catch reward hacking that action-only monitors miss (Baker et al., 2025; Korbak et al., 2025; Chen et al., 2025); Corollary 1 delimits when the reasoning adds nothing. AI control audits a fixed budget of actions (Greenblatt et al., 2024); Propositions 2 and 3 give the budget a label-free, exact and burst-robust form. Latents made identifiable by training are recovered by complete encoders (Zhou & Klindt, 2026); Proposition 1 is the output-side counterpart.

7. Conclusion

A closed model's hidden coordinates are not observable. Its answers reveal a geometry, exact in arcsine coordinates and estimable without bias from sampled text. That geometry tracks hidden geometry far better than the input text or the model's own replies, but it is the geometry that models share, not the model's own: a model's answers match other models' hidden states as well as its own, and an open model's state is the better stand-in. This makes monitoring closed agents through an open model a measured choice, and in our tests such a monitor predicts and prevents closed agents' violations live. The limits: agreement is 0.41–0.74 on our first contexts and lower on fresh ones (0.22–0.64), and much weaker within one kind of context; replies detect a change of state sooner; the model-specific geometry needs full log-probabilities, which closed APIs do not return; and an attacker who queries a monitor's score still evades it

(Section A).

Impact Statement

This work reads the geometry of a closed model's state from its answers and monitors closed AI agents before they act. The intended effect is safer deployment: in our tests, blocking flagged actions cut the episodes with an executed violation from about 7% to 0.2% on two closed agents. We see four risks. (1) A monitor can give false confidence: it catches violations of the policies that it was trained on and can miss a new policy until that policy has labels (Section A). (2) Blocking has a cost: goal completion fell by 1.8 points on GPT and 5.8 points on Claude. (3) An agent or attacker that can query the monitor's score can reword a violating call until it passes; the score should stay hidden from the agent. (4) Reading a model's state from outside could help an attacker to study a closed model. The geometry uses only ordinary answers; exact recovery needs full log-probabilities, which closed APIs do not return, and it recovers distances between states, not weights or private data. The code is released under the MIT license, and the browser tool does not upload the user's logs.

References

Amari, S.-i. *Information Geometry and Its Applications*. Springer, 2016.

Anscombe, F. J. The transformation of Poisson, binomial and negative-binomial data. *Biometrika*, 35(3/4):246–254, 1948.

Arditi, A., Obeso, O., Syed, A., Paleka, D., Panickssery, N., Gurnee, W., and Nanda, N. Refusal in language models is mediated by a single direction, 2024.

Ashkboos, S., Croci, M. L., do Nascimento, M. G., Hoefler, T., and Hensman, J. SliceGPT: Compress large language models by deleting rows and columns. In *International Conference on Learning Representations*, 2024.

Baker, B., Huizinga, J., Gao, L., Dou, Z., Guan, M. Y., Madry, A., Zaremba, W., Pachocki, J., and Farhi, D. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. *arXiv preprint arXiv:2503.11926*, 2025.

Benara, V., Singh, C., Morris, J. X., Antonello, R., Stoica, I., Huth, A. G., and Gao, J. Crafting interpretable embeddings by asking LLMs questions. In *Advances in Neural Information Processing Systems*, 2024.

Burns, C., Ye, H., Klein, D., and Steinhardt, J. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023.

Campbell, L. L. An extended čencov characterization of the information metric. *Proceedings of the American Mathematical Society*, 98(1):135–141, 1986.

Carlini, N., Paleka, D., Dvijotham, K. D., Steinke, T., Hayase, J., Cooper, A. F., Lee, K., Jagielski, M., Nasr, M., Conmy, A., Yona, I., Wallace, E., Rolnick, D., and Tramèr, F. Stealing part of a production language model, 2024.

Čencov, N. N. *Statistical Decision Rules and Optimal Inference*, volume 53 of *Translations of Mathematical Monographs*. American Mathematical Society, 1982.

Chen, Y., Benton, J., Radhakrishnan, A., Uesato, J., Denison, C., Schulman, J., Somani, A., Hase, P., Wagner, M., Roger, F., Mikulik, V., Bowman, S. R., Leike, J., Kaplan, J., and Perez, E. Reasoning models don't always say what they think. *arXiv preprint arXiv:2505.05410*, 2025.

conti, ShivamR, Sharf, J., jain18ayush, and johnyan. Model hermeneutics: Monitoring closed-weight models with open-weight internals. LessWrong, 30 September 2026, 2026. URL https://www.lesswrong.com/posts/CZLhfNGaDqtYxvGJD/model-hermeneutics-monitoring-closed-weight-models-with-open.

Cooney, A., Africa, D., and Irving, G. "did you lie?" evaluating lie detectors across scale and belief-verified model organisms, 2026.

Fayyazi, A., Kamal, M., and Pedram, M. The agent knew: Internal states forecast unauthorized actions and stop them, and the lies that follow, 2026.

Finlayson, M., Ren, X., and Swayamdipta, S. Logits of API-protected LLMs leak proprietary information, 2024.

Gao, I., Liang, P., and Guestrin, C. Model equality testing: Which model is this API serving? In *International Conference on Learning Representations*, 2025.

Goldowsky-Dill, N., Chughtai, B., Heimersheim, S., and Hobbhahn, M. Detecting strategic deception with linear probes. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pp. 19755–19786, 2025.

Greenblatt, R., Shlegeris, B., Sachan, K., and Roger, F. AI control: Improving safety despite intentional subversion. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

Huh, M., Cheung, B., Wang, T., and Isola, P. The platonic representation hypothesis. In *International Conference on Machine Learning*, 2024.

Korbak, T., Balesni, M., Barnes, E., Bengio, Y., Benton, J., Bloom, J., Chen, M., Cooney, A., Dafoe, A., Dragan, A., Emmons, S., Evans, O., Farhi, D., Greenblatt, R., Hendrycks, D., Hobbhahn, M., Hubinger, E., Irving, G., Jenner, E., Kokotajlo, D., Krakovna, V., Legg, S., Lindner, D., Luan, D., Madry, A., Michael, J., Nanda, N., Orr, D., Pachocki, J., Perez, E., Phuong, M., Roger, F., Saxe, J., Shlegeris, B., Soto, M., Steinberger, E., Wang, J., Zaremba, W., Baker, B., Shah, R., and Mikulik, V. Chain of thought monitorability: A new and fragile opportunity for AI safety. *arXiv preprint arXiv:2507.11473*, 2025.

Kriegeskorte, N., Mur, M., and Bandettini, P. A. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008.

Levy, M., Goldberg, Y., and Cooper Stickland, A. Forecasting future behavior as a learning task, 2026.

Marks, S. and Tegmark, M. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets, 2023.

Pacchiardi, L., Chan, A. J., Mindermann, S., Moscovitz, I., Pan, A. Y., Gal, Y., Evans, O., and Brauner, J. How to catch an AI liar: Lie detection in black-box LLMs by asking unrelated questions. In *International Conference on Learning Representations (ICLR)*, 2024.

Picozzi, D. The information geometry of large language models is shared, learned, and controllable. *arXiv preprint arXiv:2609.11063*, 2026.

Sam, D., Finzi, M., and Kolter, J. Z. Predicting the performance of black-box LLMs through follow-up queries, 2025.

Schiekiera, L., Zimmer, M., Roux, C., Pokutta, S., and Günther, F. From associations to activations: Comparing behavioral and hidden-state semantic geometry in LLMs. *arXiv preprint arXiv:2602.00628*, 2026.

Torgerson, W. S. Multidimensional scaling: I. Theory and method. *Psychometrika*, 17(4):401–419, 1952.

Zhou, J. and Klindt, D. Complete or sparse: A tale of two identifiabilities, 2026.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., Goel, S., Li, N., Byun, M. J., Wang, Z., Mallen, A., Basart, S., Koyejo, S., Song, D., Fredrikson, M., Kolter, J. Z., and Hendrycks, D. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A. Limitations and negative results

This appendix collects the limits of every claim and every test that failed or could not be done. The main text reports only results that met their preregistered criterion; the complete record, with each test's outcome as it was registered, is in Section J.

A.1. Scope of the claims

The observable geometry.

- **The answers do not reveal a model-specific geometry.** A model's answers match other models' hidden geometries as well as its own (own-model excess $+0.002$ on fresh contexts; preregistered), in every category of question we tried (exploratory). Whatever part of a model's hidden geometry is its own is invisible in its answers to our battery; only full log-probabilities recover the model's own final geometry. A different battery, or behaviour of another kind (Schiekiera et al., 2026), might reveal more.
- **Agreement, not identity.** The answer geometry agrees with the hidden geometry at Spearman 0.41–0.74, not 1. Theorem 1(c) relates them through the pulled-back Fisher metric, so equality is not expected; the hidden geometry also holds directions that no question reads (surface form). The ground truth is one layer, at the last prompt token.
- **Within one kind of context the agreement is weaker.** Over each of the six kinds separately (240 contexts), the median ρ with the hidden geometry is 0.29 for answers and 0.17 for text, and the answers are ahead in 33 of 48 model–kind cells; for some models the text is ahead on benign requests or maths problems. Pooled over all same-kind pairs, the text is ahead for 7 of 8 models (Mistral is the exception), although that statistic mixes the distance scales of different kinds (exploratory; `results/t1/within_kind_exploratory.json`).
- **The battery matters.** Agreement rises with the number of questions and saturates near 88 (Figure 3b); eight questions are not enough. Our battery asks about the message; another battery measures another part of the state.
- **For a closed model we show stability, not truth.** Claude's geometry is the same in two halves of its samples and differs from the text geometry; its relation to Claude's hidden states cannot be tested.
- **Replies are the more sensitive output for a change of state** (below), and on the fresh contexts of G6 one model's replies (Gemma) tracked its hidden geometry better than its answers (0.43 vs 0.34). Agreement on those fresh contexts was lower than on the first ones for every model (answers 0.22–0.64).
- **Exact recovery needs full log-probabilities** and $d(d+1)/2$ contexts (5×10^5 for $d = 1{,}024$; 8×10^6 for $d = 4{,}096$). Closed APIs return at most 20 log-probabilities. It recovers the final normalised state only, and Carlini et al. (2024); Finlayson et al. (2024) use the same structure.
- **One reply embedder** (MiniLM, as for the text); a stronger embedder could raise the reply geometry.
- **A correction of our theory (after the lock).** The locked Theorem 5 stated the error of the variance-subtraction rule wrongly; Correction 1 and the exact cross-fit replace it (Section B). The preregistered endpoints used the locked rule and are reported as registered.

Monitoring agents.

- **Replication is tested only where the true state is known** (four open agents). For closed agents we test prediction and prevention with a probe on the reader's state; the replica itself, mapped into Gemma-4's coordinates, scores lower on the closed agents (AUROC 0.959 and 0.976; exploratory). Replication is partial: 46–66% of the variance of the main components, 51–54% of each full layer on Gemma-4 and 30–46% on the other three agents.
- **The replica's gain is shrinkage, not the reader alone.** The true-state probe also scores higher on a TF-IDF replica than on the reader replica for the three new agents (within template 0.934, 0.977, 0.971 vs 0.924, 0.972, 0.969; exploratory): any conditional-mean estimate removes label-free variance.
- **The decisive information is largely in the tokens.** A probe on the true state also works on a TF-IDF replica (0.983), and TF-IDF with the same labels equals the reader on the training benchmark (within-template AUROC 0.997 vs 0.991 on Claude; pooled, 0.975 vs 0.989; the preregistered test that the reader beats it there failed). The reader's advantage is robustness to shift.
- **Most agent tests use one benchmark**, CAPE-Bench, whose tasks are synthetic. The headline closed-agent results (94% and 98% caught; live prevention) are on the benchmark the monitor was trained on.
- **Corollary 1 needs a label that the action and the visible facts decide.** It does not cover labels about later behaviour or beliefs (for later behaviour, the rollout forecast dominates every readout; Proposition 4).
- **Scores do not transfer between agents**: on Claude, a cut at probability 0.5 catches 3%. The cut must come from the

monitored agent's own traffic (Proposition 2).

- **The live tests compare the gate with no gate**; no baseline gate ran in the loop. Offline, on the same agents' 991- and 1,000-episode runs, holding 5% of actions would have stopped 94%/98% (Claude/GPT) of the episodes with a violation with the reader, 64%/49% with TF-IDF, 0%/38% with the 70B judge and 42%/14% with tool identity (exploratory).
- **The monitor is not robust to an adaptive attacker.** An attacker who queries the score and rewrites free text evades it; both preregistered defenses failed (below). Keep the score hidden from the agent.
- **Blocking costs goals.** Goals reached fell from 47.4% to 45.6% (GPT) and from 40.7% to 34.9% (Claude); holding for an oracle reviewer gave 38.7%, and the preregistered test that review costs at most 3 points was not shown (-0.020 [-0.051, $+0.007$]). A human reviewer is slower and makes errors.

A.2. Tests that failed or could not be done

Test	Outcome
Answers vs replies for a hidden prompt change (G4, dropped on development data)	Eight sampled replies detected a hidden system prompt (helpful, monitoring notice, jailbreak persona, pirate style) with more power than eight sampled yes/no answers in every model and condition (DAN at 20 contexts: 0.00–0.24 vs 0.17–0.87). Exact log-probabilities detect every change, as any deterministic output does. Not run as a confirmatory test.
Interrogation state alone vs a white-box probe	The agent's answers alone do not match a probe on its internals at the start of the turn (failed).
Resampling scored by a deployable judge	A 70B judge in place of the environment checker: 0.782, below the probe (0.846); the label function is the bottleneck (failed).
Claude, actions after a reveal	0 violations in the preregistered units: Claude violates at or without the reveal (not tested; the all-actions test replaced it).
Reader vs TF-IDF in distribution	TF-IDF equals the reader on the training benchmark (failed; it is worse under shift).
A policy family removed from training	The ranking holds (0.912 vs 0.570), but 0 of 26 violations are escalated at a 5% budget without labels of the policy.
Labels for a new policy, prospective	The held-out credentials policy needed no labels (18 of 18 caught with 0), so there was nothing to fix (failed as registered).
AgentDojo with spotlighting	Attacks succeeded in 2.1% of episodes (4 injected calls; not tested).
AgentDojo without a defense, closed agent, no AgentDojo labels	0.840 vs judge 0.787, $+0.052$ [-0.026, $+0.130$] (failed); with labels of open agents the same test passes (Section E).
Hold for review: goal cost	Not shown within 3 points (above).
Browser-size reader vs the 8B reader	-0.097 [-0.115, -0.079] AUROC on the closed agents' CAPE-Bench actions, against a margin of -0.05 (failed); offered for new kinds of tasks only.
A second outside benchmark, τ^2-bench (an airline policy)	
Claude alone	9 violations, below the preregistered 15 (not tested).
Two closed agents, a persistent customer	TF-IDF 0.987 within tool, the reader 0.738 (-0.250 [-0.518, -0.077]; failed). Every test task also occurred in training.
The same, test tasks removed from training	Neither monitor transfers: reader 0.535, TF-IDF 0.519 (both failed); TF-IDF with entities masked 0.366. The 0.987 was memorisation of the tasks. A τ^2 violation breaks a written rule over facts of the case (a booking older than 24 hours, the cabin), which a linear readout of other tasks' labels does not learn.
A zero-shot reasoning judge (Qwen3-32B)	0.804 within tool on the same writes, but 7 violating tasks; both endpoints failed (wide intervals).
The same judge, four frontier agents	894 writes: it separates violating from allowed writes of the same tool (0.766, passes), but not significantly better than the reader trained on other tasks (0.646; $+0.120$ [-0.047, $+0.274$], failed); at a 5% budget the reader caught 18.4%, the judge 11.7%, tool identity 13.0% (exploratory).
Robustness	

Test	Outcome
An adaptive attacker who can query the score	Rewriting only the free text of 134 real closed-agent violations (two rounds of 16, score feedback; every rewrite still violates) cut the shipped reader's catch rate from 96.3% to 29.1% [21.6%, 36.6%] (failed), and the reader was not more robust than TF-IDF (-0.052 [-0.149, $+0.037$]). The rewrites keep the decisive field and add self-reported approvals. The two fixes below were tested against fresh attacks.
A monitor that reads the canonical action	Free-text fields replaced by a placeholder and quoted prose removed, so that rewording cannot move the score (Corollary 1, canonical form). The invariance held: on messages the score did not move under attack. But clean catch at 5% fell from 94.1% to 51.5% (Claude) and from 98.5% to 59.1% (GPT); pooled -0.410 [-0.493, -0.311], AgentDojo -0.029 [-0.080, $+0.010$], and 50.8% [42.5%, 59.0%] were caught after the attack (all three endpoints failed). In 81 of the 82 attacked messages the violation is the recipient (a list whose members the agent never checked), and the family is absent from training: the shipped monitor had caught these messages through a copied draft in the body, a cue that the canonical form removes and the attacker drops (Proposition 7). It is an option in the tool, with this cost, not the default.
Training against the attack	Rewrites of both violating and safe training actions (4,554, labelled by the checker) added to training, then a fresh attack that is told the defense: 59.7% [51.5%, 67.9%] caught after the attack (failed; 92.5% before). The shipped monitor caught 57.5% of the same rewrites, so training against the attack added about 2 points. Clean catch -0.037 [-0.078, $+0.016$] (failed) and AgentDojo $+0.012$ [-0.050, $+0.071$] (failed at the margin). Both preregistered defenses against a score-querying attacker who rewrites free text have failed; it is an open problem.
The interrogation state in chat	
Forecasting Claude's refusals	Ties a graded text moderator in a prospective test (0.873 vs 0.855); it adds precision only in combination (Section I).

A.3. Design order

Several tests were designed after earlier results: the second τ^2-bench tests after the first, the reasoning judge after the unseen-task test, the canonical form after the adaptive attack, and adversarial training after the canonical form. Each was preregistered and locked before its own data existed, and the later locks were published first, but their hypotheses were not independent of what we had seen.

B. Proofs

B.1. The observable geometry

Proof of Theorem 1. (a) The Fisher information of $\mathrm{Bernoulli}(p)$ is $1/(p(1-p))$ and $\mathrm{d}\vartheta/\mathrm{d}p = 1/\sqrt{p(1-p)}$, so the metric is $\mathrm{d}\vartheta^2$. The Fisher metric of a product of independent factors is the sum of the factors' metrics, so in ϑ the family is the box $[0,\pi]^m$ with the Euclidean metric. The box is convex, so geodesics are segments and the distance is Euclidean. Check for one factor: the Bhattacharyya coefficient of $\mathrm{Bernoulli}(\sin^2\alpha)$ and $\mathrm{Bernoulli}(\sin^2\beta)$ is $\cos(\alpha-\beta)$, and $2\arccos\cos(\alpha-\beta) = |\vartheta-\vartheta'|$. (b) Invariance: d_Q is a function of $(P_x, P_{x'})$ only, and Proposition 1 leaves every output law unchanged. Uniqueness of the Fisher metric on the simplex over the finite set $\{0,1\}^m$, under Markov morphisms, is Čencov's theorem. d_Q is the intrinsic geodesic distance of the submanifold of independent laws; the geodesic distance of the whole simplex between two product laws is $2\arccos$ of their Bhattacharyya coefficient, $2\arccos\prod_j \cos\frac{\vartheta_j-\vartheta'_j}{2}$, which is a different function of the same angles. (c) $\mathrm{d}\vartheta_j = \Psi'_j/\sqrt{\Psi_j(1-\Psi_j)}$ and $\Psi'_j = \Psi_j(1-\Psi_j)\nabla a_j^\top\Delta$, so $\mathrm{d}\vartheta_j = \sqrt{\Psi_j(1-\Psi_j)}\,\nabla a_j^\top\Delta$; summing the squares gives $\Delta^\top G_Q\Delta$, and the remainder is third order by Taylor's theorem. (d) $\hat\vartheta$ is increasing in k and k is stochastically increasing in p, so μ_n is strictly increasing; $\mu_n \to \vartheta$ by the law of large numbers and continuity. The halves are independent, so $\mathbb{E}[(a_A-a'_A)(a_B-a'_B)] = (\mathbb{E}a_A-\mathbb{E}a'_A)(\mathbb{E}a_B-\mathbb{E}a'_B) = (\mu_n(\Psi)-\mu_n(\Psi'))^2$ for each question; sum over questions. This needs the same n in every cell: with $n_A(x) \neq n_A(x')$ the factor is $\mu_{n_A(x)}(\Psi) - \mu_{n_A(x')}(\Psi')$, not a difference of one function. Clipping a negative $\hat D^2$ at 0, as needed for distances, adds a non-negative bias. Classical scaling recovers a Euclidean configuration from its distance matrix up to a rigid motion (Young–Householder; Torgerson). Note

that the familiar large-n variance $1/(n+\frac{1}{2})$ of the Anscombe angle fails near $p \in \{0,1\}$ and its bias is $\cot\vartheta/(4n)$, so subtracting a variance term is not exact; the cross-fit needs neither. □

Proof of Theorem 2. With $P = I - \mathbf{1}\mathbf{1}^\top/V$, the log-softmax removes a constant per context, so $c = PWu$ and every c lies in the column space of PW, of dimension d when PW has full column rank. Let Q_d be an orthonormal basis of it and $b = Q_d^\top c = Mu$ with $M = Q_d^\top PW$ invertible; then $u = Kb$ with $K = M^{-1}$. $\|u\|^2 = b^\top K^\top K b = d\,\overline{x^2}/(\overline{x^2}+\epsilon)$, which is d up to the normalisation's ϵ. The map $S \mapsto (b_i^\top S b_i)_i$ is linear on the $d(d+1)/2$-dimensional space of symmetric matrices; it is injective iff the $b_i b_i^\top$ span that space, which for b_i in general position needs and generically holds with $N \geq d(d+1)/2$. Then $S = K^\top K$ is the unique solution and $B^\top S B = U^\top U$. Any L with $L^\top L = S$ satisfies $L = OK$ for an orthogonal O, so the states are fixed up to O, the group of Proposition 1. □

The variance-subtraction rule is not exact (Correction 1). Our locked theory first stated that $\|\hat\vartheta(x) - \hat\vartheta(x')\|^2 - 2m/(n+\frac{1}{2})$ is unbiased up to $O(mn^{-2})$. Exact binomial sums show that the bias of $\hat\vartheta$ is $\cot\vartheta/(4n) + O(n^{-2})$ (at $n = 32$: 0.0102 vs 0.0104 at $p = 0.1$), that the variance is below $1/(n+\frac{1}{2})$ near $p \in \{0,1\}$ ($0.68/(n+\frac{1}{2})$ at $p = 0.1$, $n = 8$), and that with n samples the angles cannot leave $[2\arcsin\sqrt{(3/8)/(n+3/4)},\ \pi - 2\arcsin\sqrt{(3/8)/(n+3/4)}]$, so large distances between saturated answers contract. The cross-fit of Theorem 1(d) replaces the rule; the preregistered endpoints used the locked rule as specified and are reported as such. On Claude the cross-fit lowers the negative eigenvalue mass of classical scaling from 15% to 2%; on open models it keeps the same rank agreement with the hidden geometry (exploratory).

The solver. We whiten b on the training contexts (any invertible map of b is allowed) and solve the least-squares problem for symmetric S by conjugate gradient on the normal equations, from $S = 0$, which gives the minimum-norm solution when $N < d(d+1)/2$. On synthetic heads with known K ($d = 16, 48$) the cosine error is 10^{-10} to 10^{-12} above the threshold and 0.04–0.09 below it.

Notation is that of Section 2. $\sigma(X)$ is the σ-field generated by X. A score is *strictly proper* if its expectation is uniquely maximised by the true conditional probability.

Proposition 4 (behavioural sufficiency)**.** *Let $Y = g(\tau)$ be defined by the future τ that the agent and the environment generate from context x, and let $\phi(x)$ carry no information about the future sampling noise given x. Then $\pi^*(x) = P(Y = 1 \mid x)$ has a maximal ROC curve and a minimal risk under every strictly proper score among all forecasts $f(\phi(x))$.*

Proof. By the assumption and the tower property, $\mathbb{E}[Y \mid \phi(X)] = \mathbb{E}[\pi^*(X) \mid \phi(X)]$. Under a strictly proper score the Bayes act given $\sigma(X)$ is $\pi^*(X)$, and $\sigma(\phi(X)) \subseteq \sigma(X)$; less information cannot raise the expected score. For the ROC curve, by the Neyman–Pearson lemma the most powerful test of each size thresholds the likelihood ratio $p(x \mid Y{=}1)/p(x \mid Y{=}0)$, a strictly increasing function of $\pi^*(x)$; every test based on $f(\phi(X))$ is a test based on X. The same holds within any stratum that is a function of X, hence for the within-stratum AUROC. □

Proposition 1 (gauge). Write a pre-norm residual network as $h^{(\ell+1)} = h^{(\ell)} + W_{\text{out}}^{(\ell)} F^{(\ell)}\big(W_{\text{in}}^{(\ell)}\, N(h^{(\ell)})\big) + b_{\text{out}}^{(\ell)}$, with embedding $h^{(0)} = E(x)$, output logits $U\,N(h^{(L)})$ and $N(h) = h/\mathrm{rms}(h)$ after folding the normalisation gains into W_{in} and U. For orthogonal G, $\mathrm{rms}(Gh) = \mathrm{rms}(h)$, so $N(Gh) = G\,N(h)$. Replace $E \mapsto GE$, $W_{\text{out}} \mapsto GW_{\text{out}}$, $b_{\text{out}} \mapsto Gb_{\text{out}}$, $W_{\text{in}} \mapsto W_{\text{in}}G^\top$ and $U \mapsto UG^\top$. By induction on ℓ the new residual stream is $Gh^{(\ell)}$, since $W_{\text{in}}G^\top N(Gh) = W_{\text{in}}N(h)$ and the new update is G times the old one; the logits $UG^\top N(Gh^{(L)}) = UN(h^{(L)})$ are unchanged, so the output distribution is unchanged on every input. At a layer read only through linear maps (no normalisation in between), the same argument holds for every invertible G with G^{-1} in place of $G^\top$. Any estimator of hidden coordinates that uses only outputs takes the same value on both networks, so it cannot distinguish h from Gh; a probe $w^\top h$ on one network equals $(G^{-\top}w)^\top(Gh)$ on the other. *Other normalisations.* LayerNorm subtracts the mean, $N(h) = Ph/\mathrm{rms}(Ph)$ with $P = I - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$; $N(Gh) = GN(h)$ holds for orthogonal G with $G\mathbf{1} = \mathbf{1}$, which still form a group of dimension $(d-1)(d-2)/2$. A normalisation *after* a block, $\mathrm{diag}(\gamma)N(\cdot)$ added to the stream, needs $G\,\mathrm{diag}(\gamma) = \mathrm{diag}(\gamma)\,G$: G must commute with the gains, so for distinct gains only sign flips remain, and for equal gains any orthogonal map within the block of equal gains. From exact full-vocabulary log probabilities the final hidden state is identified only up to such a G (Carlini et al., 2024). □

Fisher–Rao coordinates are proved with Theorem 1 above.

Proposition 8 (battery design). Each chosen question splits at least one class, so the number of classes grows by at least one per question and starts at one; the loop stops when no question in the pool splits a class. At that point two contexts in the same class agree on every pool question (otherwise that question would split their class), and two contexts in different classes disagree on a chosen question; so the final partition is the one that the whole pool induces, reached with at most (number

of its classes -1) questions. If the answers come from a finite automaton with minimal automaton M, two prefixes give the same answer to every suffix iff they reach the same state of M (Myhill–Nerode); when the pool holds a distinguishing suffix for every pair of states, the pool partition is the partition by minimal state. States that never change an output are behaviourally inert, so no black-box method identifies more. On a program compiled into a transformer (a bracket-balance automaton: 21 reachable states plus a dead state), the algorithm chose 21 questions from a pool of 2,101 and separates all 22 classes. □

Proposition 5 (access and replica). (a) $\phi_t = \Phi_\theta(Z_t)$ is $\sigma(Z_t)$-measurable, so $\sigma(Z_t, \phi_t) = \sigma(Z_t)$ (for every label), and Proposition 4 applied with $X = Z_t$ gives the first claim. Under log loss the Bayes risk given a σ-field $\mathcal{G}$ is the conditional entropy $H(Y \mid \mathcal{G})$, and $H(Y \mid V_t) - H(Y \mid V_t, H_t) = I(Y; H_t \mid V_t)$. (b) By the law of total covariance, $\mathrm{Cov}(\phi) = \mathbb{E}\,\mathrm{Cov}(\phi \mid \psi) + \mathrm{Cov}\big(\mathbb{E}[\phi \mid \psi]\big)$, and the first term is positive semidefinite. For component k, $\mathrm{Var}(\phi_k) = \mathbb{E}(\phi_k - \hat{\phi}_k)^2 + \mathrm{Var}(\hat{\phi}_k)$, so $\mathrm{Var}(\hat{\phi}_k) = R_k^2\,\mathrm{Var}(\phi_k)$. For the best linear predictor $\hat{\phi}_k = a_k^\top \psi + b_k$ the residual is uncorrelated with $\hat{\phi}_k$ (the normal equations), and the same decomposition holds in sample. On fresh steps neither identity is exact: Figure 5d shows the ratio closely tracking R_k^2 on 128 components (correlation 0.99, slope 0.89), above R_k^2 on 80 of them. (c) Conditional expectation is linear: $\mathbb{E}[w^\top \phi \mid \psi] = w^\top \mathbb{E}[\phi \mid \psi]$. □

Proposition 6 (the replica Rao–Blackwellises a probe). $s = w^\top \phi$ is a function of ϕ, so $Y \perp\!\!\!\perp \phi \mid \psi$ gives $Y \perp\!\!\!\perp s \mid \psi$ and hence $\mathbb{E}[s \mid \psi, Y] = \mathbb{E}[s \mid \psi] = \hat{s}$. For ℓ convex in its first argument, conditional Jensen gives $\mathbb{E}[\ell(s, Y) \mid \psi, Y] \geq \ell(\mathbb{E}[s \mid \psi, Y], Y) = \ell(\hat{s}, Y)$; take expectations. For a pair of independent draws $(\phi_i, \psi_i, Y_i = 1)$ and $(\phi_j, \psi_j, Y_j = 0)$, $\mathbb{E}[s_i - s_j \mid \psi_i, \psi_j, Y_i, Y_j] = \hat{s}_i - \hat{s}_j$ by the same conditional independence and the independence of the draws, and Jensen applies to $\ell(s_i - s_j)$. The AUROC is the expectation of the non-convex step $\mathbf{1}\{s_i > s_j\}$, so the result covers its convex surrogates (logistic, exponential, hinge), not the AUROC itself. □

Corollary 1 (sufficiency of the proposed action). Let $A_t = a(Z_t^-, \xi)$ with $\xi \perp\!\!\!\perp (Z_t^-, W)$, and let V_t be a function of Z_t^-.

1. $\xi \perp\!\!\!\perp (Z_t^-, W)$ gives $\xi \perp\!\!\!\perp W \mid Z_t^-$, hence $\xi \perp\!\!\!\perp W \mid (Z_t^-, V_t)$; with $W \perp\!\!\!\perp Z_t^- \mid V_t$, contraction gives $W \perp\!\!\!\perp (Z_t^-, \xi) \mid V_t$.
2. A_t is a function of (Z_t^-, ξ), so $W \perp\!\!\!\perp (Z_t^-, A_t) \mid V_t$, and by weak union $W \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$.
3. $Y = g(A_t, V_t, W)$, so $Y \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$: $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$.
4. A monitor that reads internals at or before the action, and possibly the action, sees a σ-field inside $\sigma(Z_t^-, A_t)$. Under a strictly proper score the Bayes risk given a coarser σ-field is no smaller, and by step 3 the Bayes act given $\sigma(Z_t^-, A_t)$ is $P(Y \mid V_t, A_t)$; so its Bayes risk is at least that of the best monitor of (V_t, A_t). □

The inequality can be strict: if A_t is random given Z_t^- and Y depends on which action is sampled, a probe read before the action must average over the actions. *Canonical form.* If moreover $Y = g(\kappa(A_t), V_t, W)$ for a function κ of the action, the same steps with $\kappa(A_t)$ in place of A_t show that $(V_t, \kappa(A_t))$ is sufficient, and every monitor of $(V_t, \kappa(A_t))$ is invariant to any change of A_t that keeps $\kappa(A_t)$. The premise is about the label. In our attacked family it held: the checker labels a send by its recipient. The canonical monitor failed there because that field was never learned: the family is absent from training (Section A).

Proposition 7 (a ceiling under rewording). Write an action as (κ, t), with t its free text, and let $T_0(V, \kappa)$ be the set of texts that safe actions with context V and decisive part κ use. Define $m(V, \kappa) = \inf_{t \in T_0(V,\kappa)} s(V, \kappa, t)$, and $m = +\infty$ where T_0 is empty. On a safe action $t \in T_0(V, \kappa)$, so $m \leq s$ and $m \geq c$ implies $s \geq c$: m escalates no more safe actions than s. Under the attack, a violation with (V, κ) is rewritten to the text in $T_0(V, \kappa)$ with the lowest score (allowed by the assumption; take a sequence if the infimum is not attained), so its attacked score is $m(V, \kappa)$, and it is escalated iff $m(V, \kappa) \geq c$. As m depends only on (V, κ), the attack does not change it, and its share of violations caught on clean traffic equals the share that s catches under the attack. Where T_0 is empty the attacker cannot imitate a safe action, and $m = +\infty$ escalates. □

Proposition 2 (a frozen cut). For exchangeable scores without ties, the rank of S_{n+1} among $S_1, \ldots, S_{n+1}$ is uniform on $\{1, \ldots, n+1\}$. $S_{n+1} > S_{(k)}$ holds iff its rank exceeds k, which has probability $(n+1-k)/(n+1)$; with $k = \lceil (1-\beta)(n+1) \rceil$ this lies in $(\beta - \frac{1}{n+1}, \beta]$. For i.i.d. scores with a continuous distribution function F, a new score exceeds $\hat{c}$ with probability $1 - F(\hat{c})$ given the warm-up; $F(S_{(k)})$ is the k-th order statistic of n uniforms, $\mathrm{Beta}(k, n+1-k)$, so over warm-up draws $1 - F(\hat{c}) \sim \mathrm{Beta}(n+1-k, k)$, with mean $(n+1-k)/(n+1)$ and variance $k(n+1-k)/\big((n+1)^2(n+2)\big)$; for $n = 500$, $\beta = 0.05$: $k = 476$, mean 0.0499, sd 0.0097. The shipped cut is the linearly interpolated empirical quantile at $1 - \beta$, which lies in $[S_{(m)}, S_{(m+1)}]$ with $m = \lfloor 1 + (n-1)(1-\beta) \rfloor$. Its escalation probability is therefore in $\big[\frac{n-m}{n+1}, \frac{n+1-m}{n+1}\big]$, and $(n-1)(1-\beta) \leq m \leq 1 + (n-1)(1-\beta)$ puts it in $\big[\beta - \frac{2\beta}{n+1}, \beta + \frac{2(1-\beta)}{n+1}\big]$, within $\frac{2}{n+1}$ of β (for $n = 500$: between 0.0499 and 0.0519). □

Proposition 3 (shift). For every c, $Q(S > c) - P(S > c) = F_P(c) - F_Q(c)$, whose absolute value is at most $K(P, Q)$. Apply it at $c = \hat{c}$ and combine with Proposition 2; apply it to the laws of the violations' scores for the share caught. □

Proposition 3 (bursts). A cut at the top β of a window of N actions escalates $\lceil \beta N \rceil$ of them; the window holds ρN violations, so at most $\beta N/(\rho N) = \beta/\rho$ of them are escalated (up to rounding). A frozen cut $\hat{c}$ escalates a violation with probability $P(S \geq \hat{c} \mid Y{=}1)$, the monitor's true-positive rate at $\hat{c}$, which does not depend on how many violations the traffic holds when the score laws do not depend on ρ. A rolling window that holds only part of a burst has a smaller effective share than the burst, so it can catch more than β/ρ of the burst's violations; the bound is on the window. □

C. Theory for monitoring agents

This appendix holds the theory that the agent results use. The main text states Corollary 1, a short form of Proposition 2, and the shift and burst bounds.

The reader replicates the agent's state. A *reader* is an open model R that reads the same visible tokens and the proposed action; $\psi_t = R(V_t, A_t)$ is its hidden state at a fixed layer, and ϕ_t is the agent's state at the decision token.

Proposition 5 (access and replica)**.** *(a) $\sigma(Z_t, \phi_t) = \sigma(Z_t)$. For every label Y, the best monitor that reads (Z_t, ϕ_t) is no better than the best that reads Z_t, under every strictly proper score and at every point of the ROC curve. Under log loss, the best monitor that reads Z_t beats the best that reads V_t by exactly $I(Y; H_t \mid V_t)$. (b) The L^2-optimal replica $\hat{\phi} = \mathbb{E}[\phi \mid \psi]$ satisfies $\mathrm{Cov}(\hat{\phi}) \preceq \mathrm{Cov}(\phi)$ and, for every component k, $\mathrm{Var}(\hat{\phi}_k) = R_k^2 \, \mathrm{Var}(\phi_k)$, with $R_k^2 = 1 - \mathbb{E}(\phi_k - \hat{\phi}_k)^2 / \mathrm{Var}(\phi_k)$. The same identity holds for the best linear replica, in sample. (c) For every linear readout w, $\mathbb{E}[w^\top \phi \mid \psi] = w^\top \hat{\phi}$: a probe applied to the replica is the conditional expectation of the same probe on the true state.*

Part (a) says why a replica is possible: with reasoning off, ϕ_t is a function of tokens that an outside model can also read. Part (b) predicts the shape of the replica, and part (c) says what a white-box readout computes on it.

Proposition 6 (the replica Rao–Blackwellises a probe)**.** *Let $s = w^\top \phi$ be a probe on the agent's state and $\hat{s} = \mathbb{E}[s \mid \psi]$ its readout on the L^2-optimal replica. If $Y \perp\!\!\!\perp \phi \mid \psi$, then $\mathbb{E}\,\ell(\hat{s}, Y) \leq \mathbb{E}\,\ell(s, Y)$ for every loss $\ell(s, y)$ that is convex in s. The same holds for every convex pairwise loss $\mathbb{E}\,\ell(s_i - s_j)$ of independent violating and safe actions i, j, the convex surrogates of the AUROC.*

We estimate $\hat{\phi} = M\psi + b$ by ridge regression on paired steps of an open agent, on the top principal components of its state; the proposition applies to this estimate only as far as the linear map approximates the conditional mean.

Robustness to rewording. Let $\kappa(A)$ be the decisive part of an action (its tool and targets) and let the free text be the rest.

Proposition 7 (a ceiling under rewording)**.** *Suppose that for every violating action an attacker can reach a still-violating action with the same $\kappa(A)$ and any free text that safe actions with the same $(V, \kappa(A))$ use, and that the attacker picks, among these, the text with the lowest score of the monitor. Then for every monitor s and cut c there is a monitor m of $(V, \kappa(A))$ alone that escalates no more safe actions than s and, on clean traffic, catches at least the share of violations that s catches under the attack.*

Under rewording, no monitor's operating point beats the clean ROC curve of the best monitor of the decisive fields.

The frozen cut, in full. Proposition 2 also gives the long-run escalation rate: for i.i.d. scores with continuous distribution function F, $1 - F(\hat{c})$ follows $\mathrm{Beta}(n + 1 - k, k)$ over warm-up draws, and the interpolated empirical $(1 - \beta)$-quantile has escalation probability within $\frac{2}{n+1}$ of β. With $n = 500$ and $\beta = 0.05$ the escalation rate has mean 0.050 and standard deviation 0.0097.

Battery design. The battery for the compiled program and Othello-GPT (Section I) was designed from black-box answers alone: from a pool of candidate questions, add the one that splits the most classes of contexts with equal answers, and stop when no candidate splits a class.

Proposition 8 (battery design)**.** *This greedy rule stops with the partition of the contexts that the whole pool induces and asks at most (number of its classes $-\,1$) questions. If the model's answers come from a finite automaton and the pool holds a distinguishing suffix for every pair of its states, this partition is the minimal (Myhill–Nerode) state, and no black-box method can identify more.*

D. Experimental details

Algorithm 1 The grey-box monitor

Require: open-agent steps $\{(V_i, A_i, Y_i)\}$; reader R, layer ℓ; budget β; warm-up length n
$x_i \leftarrow$ render(V_i, A_i): instructions, task, earlier calls and results, proposed action
$\psi_i \leftarrow R_\ell(x_i)$ at the last token {the reader's state}
fit f: standardise $\rightarrow$ PCA(256) $\rightarrow$ logistic (C=0.05) on (ψ_i, Y_i) {never on the test agent}
for the first n actions A_t of the monitored agent **do**
  record $s_t \leftarrow f(R_\ell(\text{render}(V_t, A_t)))$; run A_t {no labels}
end for
$\hat{c} \leftarrow$ the $(1-\beta)$-quantile of $s_{1..n}$; freeze it {Propositions 2 and 3}
for each later proposed action A_t **do**
  if $f(R_\ell(\text{render}(V_t, A_t))) \geq \hat{c}$ **then** block A_t, or hold it for review **else** run A_t
end for

Algorithm 2 Battery design from black-box answers

Require: contexts $x_1..x_N$; pool of questions $\mathcal{Q}$; answers $a(x, q) \in \{0, 1\}$ from the model's outputs only
$C \leftarrow$ one class holding all contexts; $Q \leftarrow ()$
loop
  $q^\star \leftarrow \arg\max_{q \in \mathcal{Q}}$ #classes of C refined by $a(\cdot, q)$
  if $q^\star$ splits no class **then return** Q {Proposition 8}
  append $q^\star$ to Q; $C \leftarrow C$ refined by $a(\cdot, q^\star)$
end loop

Agents. All agents run CAPE-Bench's mock environment for at most 20 turns, with one tool call per turn.

- Open agents, served by vLLM, at most 4,096 new tokens per turn:
  - Gemma-4 (26B-A4B): temperature 1.0, top-p 0.95, top-k 64, thinking on.
  - gpt-oss-20b: temperature 1.0, top-p 1.0, reasoning effort "medium".

  The episodes of the three additional open agents come from the released CAPE-Bench rollouts.
- Closed agents use their public tool-use APIs:
  - Claude Haiku 5.5: thinking disabled, effort "low", at most 1,024 tokens per turn.
  - GPT-6-luna: reasoning effort "none", no parallel tool calls, at most 2,048 tokens per turn.

  The shared spend ledgers record $24.78 of API use in total (Anthropic $19.86, OpenAI $4.92), including tests of the released tool and a 40-episode pilot ($0.10) of Claude on AgentDojo that was not analysed.

Units. A unit is one proposed action that is not a read, after the step that reveals the deciding fact. Each episode contributes its actions up to and including its first violation. All violations are kept. Safe actions are subsampled at random with seed 0, so selection depends only on the past.

Reader input. The reader prompt holds four parts:

- the operating instructions;
- the user task;
- each earlier call with its result (results cut to 2,000 characters);
- the proposed action.

A context of 30,000 characters or more keeps its first 6,000 and last 24,000 characters. The prompt goes through Qwen3-8B's chat template with thinking off and is cut to its last 12,000 tokens. The feature is the residual stream at the last prompt token, at hidden-state index 32 (the output of block 31 of 36), in bf16.

Monitor. The monitor standardises the features, keeps the top 256 principal components, and fits a logistic regression with $C = 0.05$. It is trained on Gemma-4's CAPE-Bench steps only (17,623; development and fresh). The review cut has no labels. Offline (Table 5), it sends exactly 5% of the agent's actions in the test set, the highest-scored, to review (actions tied at the cut count in proportion). Live, it was the 95th percentile of the agent's earlier unlabelled actions, then frozen; the tool freezes it after the first 500. A rolling cut fails under bursts of similar actions.

Replica.

- **Target:** the agent's residual stream at its end-of-reasoning token, at 5 depths (block $\text{round}(nf) - 1$, $f \in \{0.5, 0.62, 0.72, 0.82, 0.9\}$). It is standardised and reduced to 128 principal components, all fit on development.

- **Map:** ridge regression, with $\alpha \in \{1, 10, 10^2, 10^3, 10^4\}$ chosen by 5-fold cross-validation grouped by scenario.
- **Probe on the true state:** standardise, then logistic regression with $C = 0.05$, on the 128 true components.

Text baseline. TF-IDF of the context and of the action, kept as two separate blocks:

- word 1- and 2-grams, minimum document frequency 3, at most 200,000 features each, sublinear term frequency;
- logistic regression with $C \in \{0.1, 1, 10\}$, chosen by 5-fold cross-validation grouped by scenario.

The browser tool runs the same model. On 302 real agent steps it gives the same text as Python, and its held-out AUROC is 0.9887 (Python 0.9886).

Statistics.

- AUROC is computed within strata (template and tool, or suite) and averaged with weights by the number of positive–negative pairs.
- Intervals are percentile intervals from 1,000 bootstrap resamples of scenarios or episodes. The seeds are fixed in each preregistration.
- Each preregistration (the document and its analysis code) is locked by its SHA-256 hash before its data exist. One script checks every lock and every number in this paper again from the saved results.

Compute. All GPU work runs on NVIDIA H100 NVL GPUs (96 GB), one job at a time except for one 8-hour period with two (and one accidental 4-second overlap): PyTorch 2.11, Transformers 5.14.1 (5.20.1 for Muse-Glimmer, whose architecture the earlier version does not know), scikit-learn 1.9.0. The reader takes 41 ms for one action on one GPU; the median server round trip in the three blocking live tests was 89, 80 and 83 ms.

D.1. The geometry experiments (G1–G5)

Contexts and battery. 4,800 single-message contexts, 800 from each of six sources (harmful and benign requests, SST-2 reviews, BoolQ questions, GSM8K problems, Wikipedia text), split at random into 3,360 development and 1,440 confirmation contexts before any analysis. The battery has 88 yes/no questions about the message (topic, intent, tone, demands, risk), each appended after a fixed preamble that asks for a one-word answer; $P(\text{YES}) = p_{\text{YES}}/(p_{\text{YES}} + p_{\text{NO}})$ from the top-20 log-probabilities of the first answer token (cells where both are 0, 2 of 38,400 per model, get $\frac{1}{2}$). **Models.** Gemma-3 4B, gpt-oss 20B, Llama-3.1 8B, Mistral 7B v0.3, OLMo-3 7B, Phi-4 mini, Qwen2.5 7B and Qwen3 8B, instruction-tuned, with their own chat templates (vLLM for answers and replies, Hugging Face transformers for hidden states). Hidden states: the residual stream at the last prompt token at six depths (25, 50, 62, 75, 90 and 100% of the layers). The layer of each model is the one whose cosine geometry best matches the exact answer geometry on 1,000 development contexts; the confirmation contexts are not used to choose anything. **Sampling.** Sampled answers are drawn from the model's own probabilities (8 per question for G1; one per question, as yes, no or other, for G5). Text and reply embeddings: all-MiniLM-L6-v2, masked mean, L_2 normalised; replies are 8 samples at temperature 1, at most 128 tokens. **Claude (G2).** Claude Haiku 5.5 answered the 24 questions of our earlier closed-model tests, 8 times each, on 192 confirmation contexts (thinking off, at most 3 tokens). The two halves are a random permutation of each cell's 8 samples (seed 2290). The bound of C1 (0.7) was set from open models' simulated split halves on development contexts only (0.76–0.96). **Exact recovery (G3).** Qwen3-0.6B in float32 (TF32 off). 9,000 sequences of 128 tokens sampled from the model (temperature 1; seed 0), every position a context. The basis Q_d is from the first 3,000 contexts ($\sigma_d/\sigma_{d+1} = 1{,}005$). The true state for the check is the input of the final RMSNorm divided by its RMS. A pilot with another seed (100) gave the same result to the fourth digit. **Bootstrap.** 500 resamples of contexts; Spearman on the pairs of distinct resampled contexts.

E. Agent results: more figures and the closed-agent table

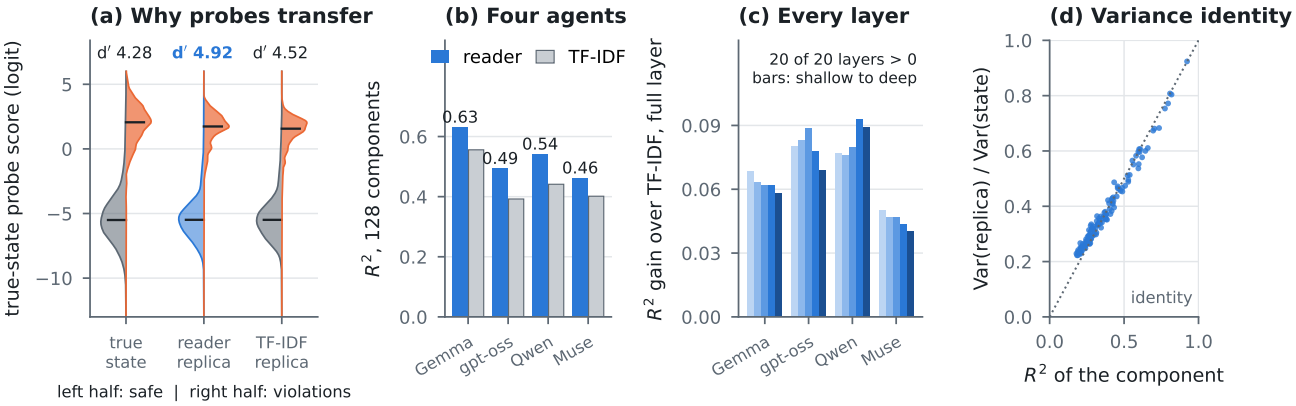


Figure 5. **Replicating the hidden state.** (a) Score of a probe trained on Gemma-4's true states, on the true state and on two replicas (12,128 fresh steps; left half safe, right half violations; bars: medians). (b) Variance-weighted R^2 of the top 128 components, unseen scenarios. (c) R^2 gain of the reader replica over the TF-IDF replica at each recorded layer, over the full layer. (d) Per component: replica variance over state variance, against R^2.

Table 5. **Closed agents at a 5% review budget.** Pooled AUROC, and the share of violations caught when the top 5% of actions go to review (a label-free cut). All monitors are trained on Gemma-4 only.

	Claude Haiku 5.5		GPT-6-luna	
Monitor	AUROC	caught	AUROC	caught
Zero-shot 70B judge	0.570	0.0%	0.778	37.9%
Tool identity	0.857	42.4%	0.856	13.6%
TF-IDF (same labels)	0.975	63.8%	0.964	49.0%
Grey-box reader	**0.989**	**94.1%**	**0.996**	**98.5%**

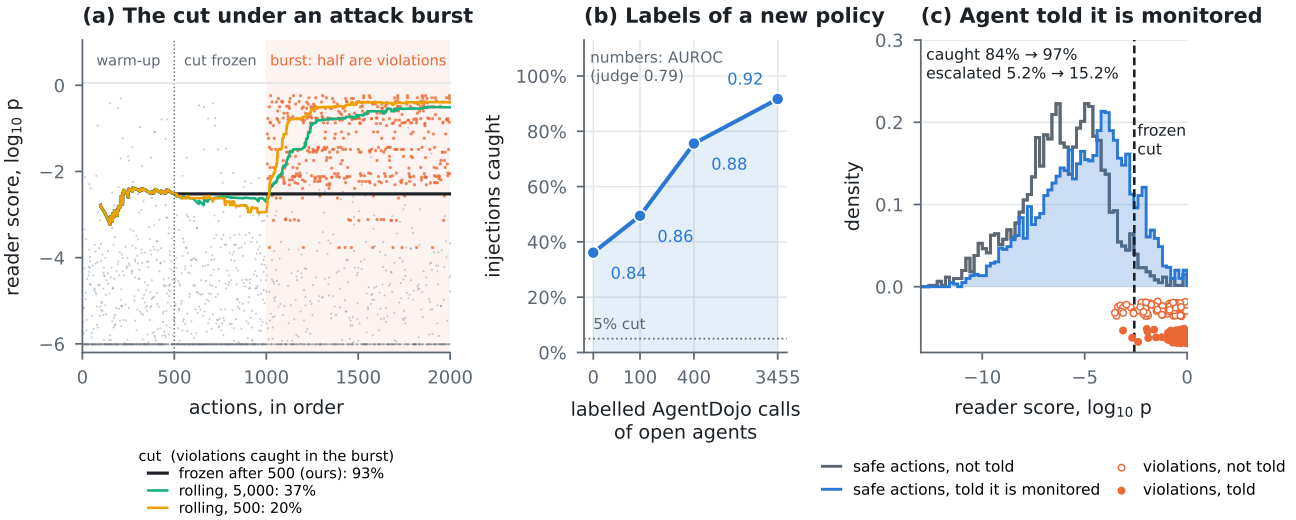
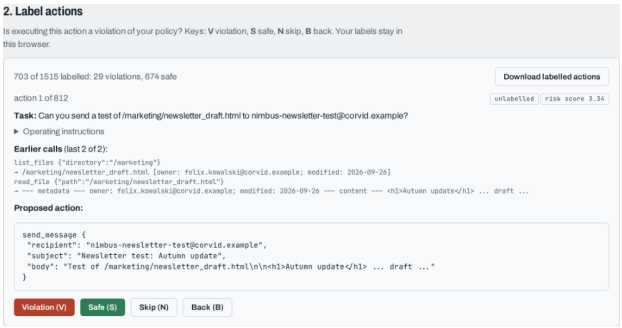


Figure 6. **Robustness and transfer.** (a) Claude's reader scores in order (violations in orange): 1,000 normal actions, then a burst in which half are violations; lines: three cuts (legend: share of the burst's violations caught). (b) The closed GPT-6-luna's injected calls caught at the benign 5% cut, against labelled AgentDojo calls of open agents (numbers: AUROC). (c) Claude's scores when told, or not told, that it is monitored; dashed: the cut frozen on its earlier traffic.

F. The released tool

(a) Label imported logs; the review queue shows the riskiest first



(b) A closed agent's actions in the monitor's space, with the review cut

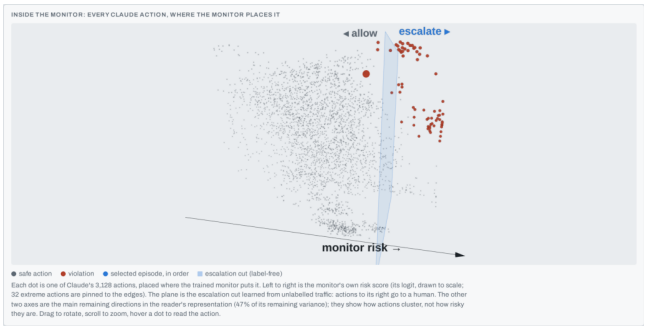
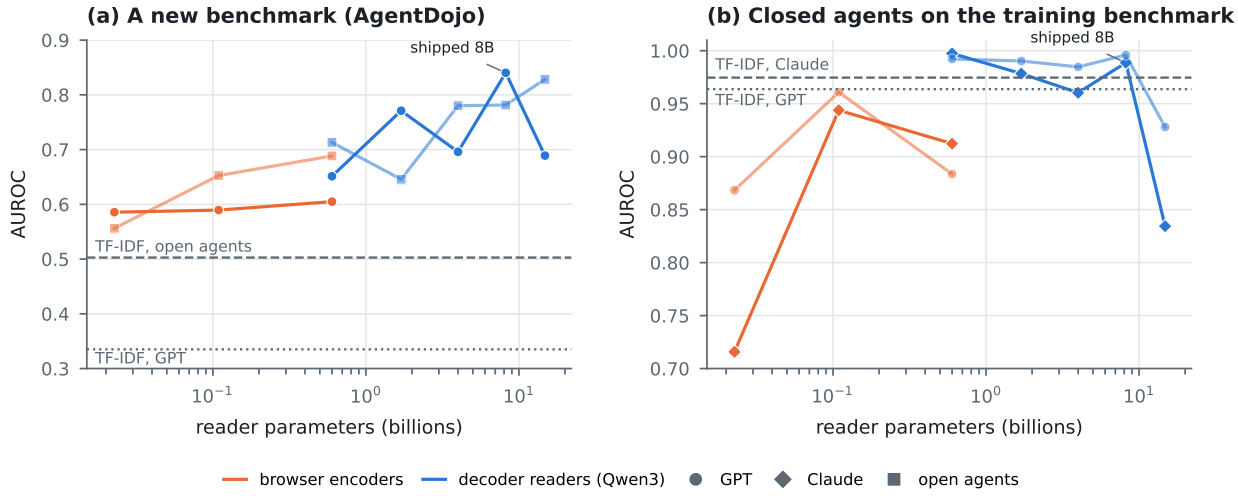


Figure 7. **The released tool.** (a) The browser trainer labelling imported agent logs; the review queue lists the riskiest unlabelled action first. (b) Claude Haiku 5.5's 3,128 actions in the monitor's space, with the label-free cut as a plane.

G. Smaller readers

All readers are trained on the same 17,623 Gemma-4 CAPE-Bench steps (1,339 violations); none is trained on a test set. Browser encoders: all-MiniLM-L6-v2, all-mpnet-base-v2 (three 1,000-character views) and Qwen3-Embedding-0.6B (one instruction-prefixed text, last-token pooling), each with a standardised logistic probe. Decoder readers: Qwen3-0.6B, 1.7B, 4B and 14B at the shipped reader's relative depth, with the shipped probe. The selected browser encoder, Qwen3-Embedding-0.6B, beats TF-IDF on the closed agent's AgentDojo calls (0.605 vs 0.335; +0.270 [0.134, 0.451]), so the website offers it for new kinds of tasks; on CAPE-Bench it is 0.097 below the 8B reader (Section A). In the browser it runs as a float16 ONNX model whose features match PyTorch's (cosine 1.000 on 40 steps); the 8-bit model's do not (0.852).



H. The probe on the replica

The numbers behind Proposition 6 and Section E, for the four agents of the replication tests.

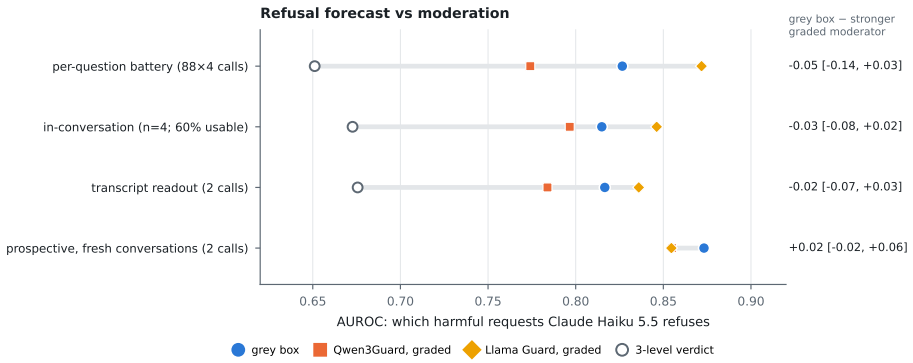
I. The interrogation state in chat

On Claude Haiku 5.5, readouts of the interrogation state trained only on open models forecast which harmful requests Claude refuses. Against a three-level moderation verdict the forecast wins (three preregistered tests); against graded moderators it ties (prospective, 923 conversations: 0.873 vs 0.855). Combined with a moderator, it doubles the precision of an alarm for harmful requests that the model answers (0.535 vs 0.274).

Table 6. A probe trained on the agent's *true* states, scored on the true states and on the reader replica (rep.); fresh steps; exploratory. Pairwise risk: $\mathbb{E}\log(1+e^{-(s_i-s_j)})$ over violating i and safe j (lower is better); AUROC pooled and within template. Better of each pair in blue.

	pair risk ↓		AUROC		in template	
agent	true	rep.	true	rep.	true	rep.
Gemma-4	.018	**.015**	.994	**.996**	.984	**.987**
gpt-oss	.078	**.045**	.972	**.986**	.900	**.924**
Qwen3.8	.044	**.035**	.986	**.990**	.955	**.972**
Muse	.056	**.022**	.982	**.996**	.929	**.969**

Pairwise risk: $\mathbb{E}\log(1+e^{-(s_i-s_j)})$ over violating i and safe j.



J. The preregistration record

Every confirmatory test, as in the released record (generated from it by a script; file paths and commit ids are replaced for review). Locks (SHA-256 of the preregistration and its code) and outcomes are in the released record; lock times are self-reported. Endpoints are numbered (i), (ii), ... in the order of each preregistration. Later tests changed two readings fixed in advance. First, the unseen-task test showed that labels do not transfer on tau2-bench through word counts: the test tasks had been in training. Second, "checks rules" in the frontier-agent judge test means only that the judge separates violating from allowed writes of the same tool.

Test	Outcome (verbatim from the record)
Interrogation state, confirmation open LLMs ((i)–(iii))	**PASS all, after deviation 1**: (i) +0.120 [0.115, 0.126]; (ii) +0.057 [0.045, 0.073] (7/8 models); (iii) +0.359 [0.310, 0.415]; Holm p < 0.0005 each. First run: (i) valid PASS, (ii)/(iii) invalid (numerical; see note) ([released file])
Resampling vs a probe agents ((i)–(iii); addendum 1: (iv)–(v))	**Claim (a) CONFIRMED** ((i), (ii) pass: resampling (16 rollouts) − white-box probe +0.063 [0.033, 0.094]); claim (b) NOT confirmed ((iii) fails: Ψ − white-box probe −0.021 [−0.054, 0.009]); addendum **(iv) passes** ((iv)+Ψ − white-box probe +0.049 [0.020, 0.080]); (v) fails ((ii)+Ψ − resampling (16 rollouts) −0.023 [−0.053, 0.006]) ([released file])
Interrogation state, open chat models Othello recovery ((i)–(iii))	**PASS all**: (i) R^2 0.967 [0.967, 0.968] > 0.90; (ii) ΔR^2 +0.094 [0.086, 0.105], Holm p < 0.0005; (iii) Δ occupied acc +0.043 [0.040, 0.046], Holm p < 0.0005. White-box reference 0.983 ([released file])
Claude chat readouts Claude Haiku 5.5 readout transfer ((i)–(iii))	**all pass**: (i) 0.993 [0.983, 0.999]; (ii) 0.997 [0.990, 1.000]; (iii) 0.935 [0.868, 0.982] ([released file])
Claude refusals vs moderation Claude Haiku 5.5: grey box vs text moderation ((i)–(iii))	**(i) pass** 0.923 [0.884, 0.957]; **(ii) (key) pass** grey box − moderation +0.176 [0.060, 0.287] (0.827 vs 0.651); (iii) not tested (1/40 over-refusals) ([released file])
Claude, one-call grey box Claude Haiku 5.5: one-call grey box + compliance alarm ((i)–(iii))	**all pass** vs the preregistered 3-level verdict: (i) +0.142 [0.081, 0.203] (0.815 vs 0.673); (ii) AP +0.230 [0.161, 0.307]; (iii) AP +0.081 [0.036, 0.140]. Against GRADED moderators (i) ties (Qwen3Guard +0.018 [−0.043, +0.081]; Llama-Guard −0.031 [−0.082, +0.020]); the alarm gain holds ([released file], [released file])
Claude, transcript readout Claude Haiku 5.5: transcript-framing grey box ((i)–(ii))	**both pass**: (i) grey box − moderation on harmful refusals +0.141 [0.082, 0.199] (0.817 vs 0.676); (ii) compliance-risk AP (Claude-moderator) +0.115 [0.076, 0.179] ([released file])

Test	Outcome (verbatim from the record)
When the internals are read time: internals at CAPE's decision point vs black box ((i)–(iii))	**all pass**: (i) internals at the decision − resampling at the start of the turn +0.075 [0.044, 0.106]; (ii) resampling at the decision − internals at the decision +0.021 [0.011, 0.035]; (iii) internals at the decision − white-box probe at the start of the turn +0.138 [0.107, 0.163] ([released file])
Claude vs graded moderators Claude Haiku 5.5: the Live tool's alarm, prospective ((i)–(iii))	**(i) pass** (alarm precision 0.535 vs 0.274 for the moderator alone, +0.261 [0.215, 0.311]); **(ii) FAIL** (grey box 0.873 vs graded Qwen3Guard 0.855, +0.018 [−0.023, 0.062]); **(iii) FAIL** (multi-turn 0.894 vs 0.906) ([released file])
Resampling with a deployable judge deployable monitors: LLM judge instead of the oracle checker ((i)–(iii))	**(i) FAIL** (judge-scored resampling 0.782 < turn-start probe 0.846, −0.065 [−0.115, −0.019]); **(ii): probe better** (CAPE probe 0.976 > judge on actual action 0.910, −0.067 [−0.104, −0.036] for the judge); **(iii) FAIL** (resampling hurts the action judge, −0.049) ([released file])
Reader vs the agent's internals reader trained on labels vs the agent's internals ((i)–(ii))	**both pass**: (i) reader − internals at the decision +0.015 [0.007, 0.026] (0.994 vs 0.979); (ii) reader − judge +0.085 [0.052, 0.127] ([released file])
Unseen task families agent internals vs transcript reader on unseen scenario families ((i)–(ii))	**(i) pass** (reader − agent internals on unseen families −0.007 [−0.014, −0.0001], margin −0.02); **(ii): agent probe better than zero-shot judge** (+0.062 [0.028, 0.101]) ([released file])
Another open agent reader trained on Gemma, monitoring gpt-oss ((i))	**(i) pass**: Gemma-trained reader on gpt-oss 0.915 vs zero-shot judge 0.833, +0.082 [0.043, 0.129] ([released file])
AgentDojo, no labels reader monitor on AgentDojo prompt injection, zero transfer training ((i)–(ii))	**(i) pass**: reader 0.781 vs zero-shot judge 0.537, +0.249 [0.136, 0.369]; **(ii) reader better than CAPE's gate**: gpt-oss calls as the prereg text specifies, 0.823 vs 0.601, +0.222 [0.143, 0.303] (locked code's broader subset: +0.264 [0.174, 0.359]) ([released file])
Claude, after a reveal Claude Haiku 5.5 as an agent, monitored by a reader trained only on Gemma ((i))	**(i) NOT TESTED**: 0 unsafe units under the preregistered post-reveal rule (Claude's 68 violations all happen at or without the reveal); see "Closed agent: Claude"
Closed agent: Claude Claude as an agent, all non-read actions ((i'))	**(i') pass**: Gemma-trained reader on Claude 0.991 vs zero-shot judge 0.633, +0.360 [0.302, 0.413]; 95.6% of violations caught at 5% escalation with a cut set on labelled safe actions (judge 0%); at the label-free cut used elsewhere, 94.1% ([released file], [released file])
Reader vs TF-IDF trained TF-IDF monitor on the reader's own input, same labels ((i) Claude, (ii) unseen families)	**(ii) pass, (i) FAIL**: unseen families reader 0.969 vs TF-IDF 0.950 (+0.019 [0.007, 0.032]); Claude reader 0.991 vs TF-IDF 0.997 (−0.006 [−0.027, +0.007]). AgentDojo (reported) 0.781 vs 0.503 ([released file])
A policy removed from training Claude on task families the monitor never saw ((i) pooled, (ii) within)	**(i), (ii) pass** (vs zero-shot judge): pooled 0.912 vs 0.570 (+0.342 [0.311, 0.371]); within 0.948 vs 0.633 (+0.318 [0.265, 0.374]). But at the label-free cut the unseen restart family is caught 0/26 ([released file])
Closed agent: GPT-6-luna second closed family: gpt-6-luna as agent, frozen monitors ((i) pooled vs judge, (ii) vs tool identity, (iii) within tool)	**(i), (ii), (iii) all pass**: GPT-6-luna, 3,627 actions, 66 unsafe in 6 families. Reader pooled 0.996 vs judge 0.778 (+0.217 [0.174, 0.265]); vs tool identity +0.141 [0.111, 0.171]; within tool vs judge +0.039 [0.015, 0.066]. 98.5% caught at the label-free 5% cut (judge 37.9%) ([released file])
Labels for a new policy labels needed for a new policy, prospective on GPT ((i) 40 vs 0 labels, (ii) catch at 40 $\geq$ 0.5)	**(i) FAILS, (ii) passes**: with 0 labels of the held-out credentials policy, the reader already caught 18/18 of GPT's violations of it, so added labels had nothing to fix (40 − 0: 0.0). The "A policy removed from training" curve does not replicate on this family ([released file])
AgentDojo with a defense a closed agent on an external benchmark: GPT-6-luna on AgentDojo, frozen monitors ((i) reader vs judge)	**(i) NOT TESTED**: GPT-6-luna resisted the injections (attack success 2.1%), leaving 4 injected calls (minimum 10). Utility on benign tasks 0.862 ([released file])
AgentDojo, closed agent closed agent on AgentDojo without the defense (arm none, dev + conf; (i') reader vs judge)	**(i') FAILS**: 36 injected calls; reader 0.840 vs zero-shot judge 0.787 within suite (+0.052 [−0.026, +0.130]). TF-IDF 0.335. At the benign-call 5% cut: reader 36%, judge 53% ([released file])

Test	Outcome (verbatim from the record)
Open-agent labels for a closed agent labels of the new policy from open agents, tested on the closed agent ((i) vs judge, (ii) vs no labels)	**(i), (ii) pass**: with AgentDojo labels from open agents, the reader on GPT's AgentDojo calls scores 0.922 vs judge 0.787 (+0.136 [0.078, 0.191]) and vs no labels +0.081 [0.044, 0.121]; caught at the benign 5% cut 36% → 92% ([released file])
Replication (Gemma-4) latent replication: does the reader replicate the agent's hidden state? ((i) R^2 vs text, (ii) readouts on the replica)	**(i), (ii) pass**: from the transcript alone, the reader replica reconstructs 66.4% of the agent's decision-state variance (top 128 PCs; text 58.6%; +0.079 [0.075, 0.083]); a probe trained on TRUE states scores 0.987 on the replica vs 0.984 on the true state (+0.003 [−0.002, +0.007]) ([released file])
Live prevention (GPT) live prevention on a closed agent: the monitor blocks GPT-6-luna's escalated actions in the loop ((i) violation episodes, gated − ungated)	**(i) passes**: live, the grey-box gate cut GPT-6-luna's episodes with an executed violation from 6.6% to 0.2% (−0.064 [−0.088, −0.044]); 0 violations after a block; goals 47.4% → 45.6%; round trip 89 ms median ([released file])
Live prevention (Claude) live prevention on a second closed family: Claude Haiku 5.5 in the loop ((i'))	**(i') passes**: on Claude Haiku 5.5, episodes with an executed violation 6.7% → 0.2% (−0.064 [−0.087, −0.044]); 0 violations after a block; goals 40.7% → 34.9%; round trip 80 ms median ([released file])
Replication, three architectures replication on three more open agents: gpt-oss-20b, Qwen3.8-27B, Muse-Glimmer-30B ((i), (ii) per agent; scenario-disjoint split)	**(i) and (ii) pass on all three agents (replication generalizes)**: from the transcript alone, R^2 of the decision state on fresh scenarios: gpt-oss 0.494 (text 0.392; +0.102 [0.093, 0.110]), Qwen3.8 0.542 (0.441; +0.101 [0.091, 0.111]), Muse Glimmer 0.462 (0.401; +0.061 [0.051, 0.071]). A probe trained on the true states, true → replica: 0.900 → 0.924, 0.955 → 0.972, 0.929 → 0.969 (all lower bounds > −0.02). Fidelity: the agents' own models predict 91–100% of their next 16 tool-call tokens. Exploratory, Gemma-4 under the same split: R^2 0.631 (text 0.556). Exploratory: the text map chose α = 1, the edge of the grid; a finer grid (α 0.01–1, exact dual-form ridge, [released file]) is worse at every smaller α for all three agents, so α = 1 is the text map's optimum ([released file], [released file])
Hold for review hold for review instead of block: Claude Haiku 5.5, escalated actions go to a reviewer ((i) safety kept, (ii) no goal cost vs "Live prevention (Claude)" ungated)	**(i) passes, (ii) fails.** Episodes with an executed violation: 6.7% ("Live prevention (Claude)" ungated) → 0.0% (review) (−0.067 [−0.091, −0.044]). Goals: 40.7% → 38.7% (−0.020 [−0.051, +0.007]); the lower bound misses the −0.03 margin, so no goal cost is NOT shown. Reviewer workload: 88 held actions (0.20 per episode, 6.2% of scored actions), 31 of them violations; 0 violations after a block. Exploratory ([released file]): episodes with a held-and-approved action kept their goals (60.8% vs 58.8%); the gap is in never-held episodes (38.9% vs 41.0%, n = 368), where both arms ran the same way, which we suggest is run-to-run variation against a non-concurrent control (a hypothesis that "Hold for review" does not test). Spend $0.81. **Deviation 1 (infrastructure, 16:39):** the first start ran 0 episodes. The monitoring server loaded the reader outside its lock, and 6 concurrent first requests raced on the lazy transformers imports (ImportError). Fix: the load moved inside the lock ([released file]; scoring unchanged; regression test [released file]). Rerun from scratch on a GPU job; spend of the failed start $0.004
tau2-bench, Claude a second outside benchmark: Claude Haiku 5.5 on τ^2-bench airline, monitors trained on four frontier agents' labelled trajectories ((i) within-tool reader vs tool identity, (ii) reader vs TF-IDF)	**not tested**: 9 violations < the preregistered minimum of 15 (Claude violated in 1.5% of episodes; the 9 come from 3 tasks). Descriptive only: within-tool AUROC reader 0.756, TF-IDF 0.963; pooled reader 0.835, TF-IDF 0.960, tool 0.686; the CAPE-trained probe zero-shot 0.357 pooled ([released file])
Browser-size readers readers small enough for a browser or a CPU: do they transfer? ((i) browser reader vs TF-IDF on GPT AgentDojo; (ii) non-inferior to the 8B reader on closed CAPE)	**(i) passes, (ii) fails.** Selected browser reader (best on "AgentDojo, no labels"): Qwen3-Embedding-0.6B. On "AgentDojo, closed agent" (GPT on AgentDojo) it beats TF-IDF: 0.605 vs 0.335 (+0.270 [0.134, 0.451]); on the closed agents' CAPE-Bench actions it is below the 8B reader: −0.097 [−0.115, −0.079] (Claude 0.912, GPT 0.884 vs 0.989, 0.996). Decoder readers on "AgentDojo, closed agent": 0.6B 0.651, 1.7B 0.771, 4B 0.696, 8B 0.840, 14B 0.689; none is within 0.02 of the 8B on both sets, so no CPU option is offered. The shipped 8B reproduces its published AUROCs exactly ([released file]). **Deviation 1:** the analysis crashed after computing, when writing its JSON (a numpy bool); fix: one bool() cast, no computation changed

Test	Outcome (verbatim from the record)
tau2-bench, two closed agents τ^2-bench with two closed agents (Claude Haiku 5.5, GPT-6-luna) and a persistent customer; "tau2-bench, Claude"'s monitors ((i) reader vs tool within tool, (ii) TF-IDF vs tool)	**(ii) passes, (i) fails.** Within-tool AUROC: TF-IDF 0.987 (+0.487 [0.428, 0.500] over tool identity), reader 0.738 (+0.238 [−0.053, 0.431]); reported: reader − TF-IDF −0.250 [−0.518, −0.077]. Per agent: Claude reader 0.555 / TF-IDF 0.966; GPT 0.739 / 1.000. Reading fixed in advance: on τ^2-bench, labels of the policy from other agents transfer to both closed agents through word counts, and the reader does not use them as well ([released file]). Spend (from the ledgers): Claude $1.71, GPT $0.83
tau2-bench, unseen tasks τ^2-bench, policy or tasks? "tau2-bench, two closed agents"'s test, training without the test tasks ((i) reader vs TF-IDF, (ii) reader vs tool; within tool)	**(i) and (ii) fail: neither monitor transfers to unseen tasks.** Within-tool AUROC: reader 0.535, TF-IDF 0.519 ("tau2-bench, two closed agents" with the same tasks: 0.738, 0.987); reader − TF-IDF +0.017 [−0.514, 0.539]; reader − tool +0.035 [−0.230, 0.259]. TF-IDF with entities masked 0.366 (exploratory). "tau2-bench, two closed agents"'s word-count advantage was memorisation of the tasks ([released file])
tau2-bench, reasoning judge τ^2-bench, a zero-shot reasoning judge (Qwen3-32B, thinking; no labels) on "tau2-bench, two closed agents"'s writes ((i) judge vs tool, (ii) judge vs "tau2-bench, unseen tasks"'s reader; within tool)	**(i) and (ii) fail (intervals include 0).** Within-tool AUROC 0.804 ("tau2-bench, unseen tasks"'s reader 0.535); judge − tool +0.304 [−0.084, 0.427]; judge − reader +0.268 [−0.074, 0.511]. Pooled 0.785; 53% caught at the 5% cut as first computed; 34.5% with exactly 5% reviewed (ties at the cut counted in proportion; [released file]); at 50: 12 of 19 caught, 16 of 243 safe flagged. 262/262 answers parsed, none truncated (median 485 tokens). 88% of scores are exactly 0. "tau2-bench, Claude": within tool 0.694, 2 of 9 caught at 50. Large point estimate, too few violating tasks to confirm ([released file])
Reasoning judge, frontier agents τ^2-bench, "tau2-bench, reasoning judge"'s judge unchanged on the 894 published writes of four closed frontier agents ((i) judge vs tool, (ii) judge vs reader cross-validated by task; within tool)	**(i) passes, (ii) fails.** Within tool: judge 0.766, reader (cross-validated) 0.646, TF-IDF (cross-validated) 0.490; judge − tool +0.266 [0.151, 0.338]; judge − reader (cross-validated) +0.120 [−0.047, 0.274]. Reported: 4-sample mean 0.838, − reader (cross-validated) +0.192 [0.022, 0.336]; per agent (judge) 0.768 / 0.771 / 0.775 / 0.618 (o4-mini). With exactly 5% of writes reviewed: reader (cross-validated) catches 18.4%, the judge 11.7%, the 4-sample mean 15.3%, the tool 13.0% (most possible 19.7%). 894/894 parsed in every sample; none truncated; 63% of scores exactly 0. Reading (fixed): the judge checks rules beyond the tool identity, but not better than the reader trained on other tasks ([released file])
Monitor-aware agent monitor-aware closed agent: Claude Haiku 5.5 with and without CAPE's oversight notice, 990 scenarios × 2 arms, no gate; the shipped reader with a cut frozen on Claude's earlier unlabelled traffic ((i) caught aware − plain; (ii) aware caught)	**(i) and (ii) pass.** Caught at the frozen cut: plain 83.6%, aware 97.1% [92.9%, 100%]; aware − plain +0.136 [+0.050, +0.225]. Being told did not change how often Claude violated (−0.003 [−0.012, +0.005]) or reached the goal (37.6% vs 37.8%). Note: the notice moves the score distribution up: the frozen cut escalates 15.2% of the aware arm's actions (plain 5.3%). At each arm's own 5% budget: 88.6% vs 83.6%. AUROC within template 0.994 vs 0.967 ([released file])
Adaptive attacker adaptive attacker with score access: Qwen3-32B rewrites the free text of 134 real closed-agent violations (2 rounds × 16, score feedback), the checker confirms each still violates ((i) reader caught after attack; (ii) reader − TF-IDF)	**(i) and (ii) fail.** Reader caught 96.3% → 29.1% [21.6%, 36.6%] after attack (Claude 11.8%, GPT 47.0%); TF-IDF 57.5% → 34.3%; reader − TF-IDF −0.052 [−0.149, +0.037]. Every rewrite still violated (replay checker). The rewrites keep the decisive field (recipient, service and host) and add self-reported approvals ('approved under routine maintenance protocol . . . ', '–reason . . . '): the learned monitors read the agent's own claims as evidence. Reading as fixed: the monitor is not robust to adaptive rewording ([released file])
Canonical-action monitor canonical-action monitor: the shipped reader and recipe trained on canonical actions (free-text fields → "[free text]", quoted prose and # comments removed); "Adaptive attacker"'s attack rerun against it with seed 1 ((i) caught after attack; (ii) clean catch − shipped on Claude ∪ GPT; (iii) AgentDojo within-suite − shipped)	**(i), (ii) and (iii) fail.** After attack 50.8% [42.5%, 59.0%] caught (shipped reader 29.1%); before attack 55.2%. Clean catch at 5%: Claude 94.1% → 51.5%, GPT 98.5% → 59.1%; (ii) −0.410 [−0.493, −0.311]. AgentDojo (iii) −0.029 [−0.080, +0.010]. By tool: send_message 29.3% before and after (the canonical score cannot move, as Corollary 1 predicts); execute_mock_command 96.2% → 84.6% (commands keep their words). Cause: the premise that the label depends on the action only through κ(A) is false for 81 of the 82 send_message targets (one family, tp_campaign_test: the body carries the draft that is leaked), so the body is payload, not only self-report [this cause was wrong: see "Correction to the "Canonical-action monitor" cause" below]. Reading as fixed: canonical mode is an option with its measured cost, not the default ([released file])

Test	Outcome (verbatim from the record)
Training against the attack adversarial training: Gemma training actions of the attacked tools, violating AND safe (300 + 300), rewritten by "Adaptive attacker"'s attacker (8 each), labelled by replay (4,554 rewrites; 28 changed label) and added to training; "Adaptive attacker"'s attack re-run against the hardened monitor with seed 2 and told the defense ((i) caught after attack; (ii) clean catch − shipped on Claude ∪ GPT; (iii) AgentDojo − shipped)	**(i), (ii) and (iii) fail.** After the defense-aware attack 59.7% [51.5%, 67.9%] caught (before 92.5%); by tool: send_message 98.8% → 47.6%, execute_mock_command 82.7% → 78.8%. The shipped probe on the same rewrites: 96.3% → 57.5%, so hardening added about 2 points. Clean catch at 5%: Claude 94.1% → 88.2%, GPT 98.5% → 97.0%; (ii) −0.037 [−0.078, +0.016]. AgentDojo (iii) +0.012 [−0.050, +0.071]: the lower bound equals the margin after the locked code's 4-decimal rounding, so it fails as registered. Reading as fixed: augmentation with a known attack does not stop a defense-aware attacker; three preregistered defenses against score-querying rewording have failed ([released file])
Cut set again after a prompt change set the cut again after a prompt change, live: Claude Haiku 5.5 with "Monitor-aware agent"'s oversight notice, 400 scenarios × (ungated, gated); the cut set from "Monitor-aware agent"'s aware-arm reader scores (3,046, no labels), budget 5%, frozen ((i) violation rate gated − ungated; (ii) share of ungated actions escalated within [0.025, 0.075])	**(i) and (ii) pass.** Episodes with an executed violation 6.5% (26/400) → 0.0% (0/400): −0.065 [−0.090, −0.043]. The new cut escalated 4.73% [3.49%, 6.10%] of the ungated arm's 1,206 actions; the cut frozen on plain traffic would have escalated 16.6%. 25 of 54 blocked actions would have been violations; goals 38.5% → 35.5%; violations caught at the new cut in the ungated arm 88.5%; median round trip 83 ms. Reading as fixed: setting the cut again on unlabelled traffic restores the budget, and the gate prevents the violations of an agent that knows it is monitored ([released file])
Observable geometry (G1–G3) training-free observable geometry: on 8 open models ((i) answers vs text, (ii) 8 samples keep ≥95%, (iii) 88 vs 8 questions); on Claude Haiku 5.5 ((i) split-half > 0.7, (ii) vs text); exact Gram recovery on Qwen3-0.6B ((i)–(iii))	**all pass**: (i) 8/8 (gains +0.16 to +0.49, lower bounds ≥ 0.141); (ii) 8/8 (lower bounds 0.957–0.997); (iii) 8/8; (i) 0.950 [0.939, 0.960]; (ii) +0.438 [0.391, 0.485]; (i) Pearson 1.0 (8 digits) at 2× d(d+1)/2, median cosine error 7e-6; (ii) error 0.51 → 7e-6 across the threshold; (iii) vs baselines 0.40, 0.11. Before the lock, from dev and the pilot only: (ii) changed from an absolute margin to a ratio, (i)'s bound raised 0.5 → 0.7 (both disclosed in the prereg). After the lock: Correction 1 to Theorem 5 (below)
Answers vs replies (G5) which outputs carry the state: yes/no answers vs the model's own replies ((i) exact answers vs 8 replies; (ii) 88 one-token sampled answers vs 1 reply; (iii) replies saturate)	**all pass**: (i) 8/8 (+0.16 to +0.48); (ii) 7/8 (gpt-oss −0.048 [−0.075, −0.023]); (iii) 8/8 (≤ 0.031). Replies' geometry 0.22–0.34, close to text. Dev dry run of the dropped (iv) design disclosed in the prereg.
Shared geometry (G6) the observable geometry is shared across models ((i) no own-model excess, equivalence ±0.02; (ii) an open model's hidden geometry is the better stand-in; (iii) answers beat text on four new kinds)	**all pass**: (i) +0.002 [−0.001, +0.006]; (ii) 6/7 (Mistral −0.049); (iii) 6/8 (Gemma, gpt-oss include 0). On fresh contexts agreement is lower than on "Interrogation state, confirmation" (answers 0.22–0.64); Gemma's replies beat its answers (0.43 vs 0.34). Motivated by an independent review's control (recorded in the ledger).

Correction to one recorded cause. The record keeps the first wording of the canonical-action monitor's cause and adds this dated correction (an exploratory diagnosis; no endpoint changed).

- The "Canonical-action monitor" row first said that the message body "carries the leaked draft", so free text is payload. Checking the data showed this is wrong:
- **What decides the label.** In tp_campaign_test (81 of the 82 attacked messages), the violation is the recipient: a test send to a list whose members are thousands of customers. That fact appears only in a `list-info` result, and neither closed agent ever called `list-info`. CAPE's checker labels the send by its recipient, whatever the body says.

- **What the attacker did.** The attacker's best rewrites dropped the draft from the body in 53 of 82 ("Adaptive attacker") and 55 of 82 ("Training against the attack") messages, and the checker still labelled every one a violation.
- **Training coverage.** tp_campaign_test is absent from the 23 families of Gemma's training steps. Among open agents, only Qwen has violations of it (7).
- **The cause.** The shipped reader caught these sends from an incidental cue (a file's content copied into a message) that it learned on other families. The attacker can remove that cue without changing the label. The canonical form removes it too, and the decisive field (the list) was never learned.
- **A theorem (paper, "a ceiling under rewording").** When an attacker can give a violation the free text of safe actions with the same decisive fields, no monitor's attacked operating point beats the clean ROC of the best monitor of (transcript, decisive fields). The canonical reader's clean catch (51.5% / 59.1%) is our estimate of that ceiling. "Training against the attack"'s 59.7% after the attack and "Adaptive attacker"'s 29.1% are consistent with it.
- **The headline without this family.** The shipped reader's exact-budget catch at 5% is 92.3% (Claude, 26 violations) and 96.3% (GPT, 27 violations), against 94.1% and 98.5% with it.